CHAPTER 2

Exercise 2-1

with n input variables for logic functions 2^{2^n}.

2^{2^n} for 1 = 4

2^{2^n} for 2 = 16

2^{2^n} for 3 = 256

Exercise 2-2

Construct the truth table for the circuit in Fig 2-62

4

First gate = $\bar{Y}$

Second gate ⇒ $\overline{X \cdot \bar{Y}}$

Third gates ⇒ $\overline{X \cdot \bar{Y} \cdot X}$, $\overline{X \cdot \bar{Y} \cdot \bar{Y}}$

Fourth gates ⇒ $\overline{X \cdot \bar{Y} \cdot X} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z}$, $\overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z}$

Final gate (F) ⇒ $\overline{X \cdot \bar{Y} \cdot X} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z}$

$F = \overline{X \cdot \bar{Y} \cdot X} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z}$

$F = \overline{X \cdot \bar{Y} \cdot X} \cdot \overline{X \cdot \bar{Y} \cdot \bar{Y} \cdot Z} + X \cdot \bar{Y} \cdot \bar{Y} \cdot Z$

$F = [\overline{X \cdot \bar{Y}} + \bar{X}] \cdot [\overline{X \cdot \bar{Y}} + \bar{Y}] Z + [\overline{X \cdot \bar{Y}} + \bar{Y}] \cdot Z$

$F = [\bar{X} + Y] \cdot [X \cdot \bar{Y} + Y] Z + [X \cdot \bar{Y} + Y] \cdot Z$

$F = [\bar{X} + Y] \cdot [0] + [X + Y] \cdot Z$

$F = 0 + [X + Y] \cdot Z$

$F = X \cdot Z + Y \cdot Z$

$(X \cdot \bar{Y} + Y) = (X + Y)$

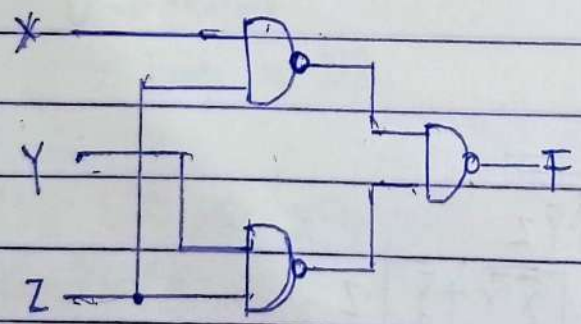
$(X \cdot Y + \bar{Y}) = (X + \bar{Y})$

X	Y	Z	F	
0	0	0	0	$F = XZ + YZ \Rightarrow (0)(0) + (0)(0) = 0$
0	0	1	0	$F = XZ + YZ \Rightarrow (0)(1) + (0)(1) = 0$
0	1	0	0	$F = XZ + YZ \Rightarrow (0)(0) + (1)(0) = 0$
0	1	1	1	$F = XZ + YZ \Rightarrow (0)(1) + (1)(1) = 1$
1	0	0	0	$F = XZ + YZ \Rightarrow (1)(0) + (0)(0) = 0$
1	0	1	0	$F = XZ + YZ \Rightarrow (1)(0) + (0)(1) = 0$
1	1	0	0	$F = XZ + YZ \Rightarrow (1)(0) + (1)(0) = 0$
1	1	1	1	$F = XZ + YZ \Rightarrow (1)(1) + (1)(1) = 1$

Logic Circuit $F = Z(X+Y)$

Using a minimum number of nand gates.
more than one but less than 4??

$X \cdot Z + \bar{X} \cdot Z$



Exercise 2.3: Simplify the following

a) $X + \bar{Y} + \bar{X} \cdot Y + (X + \bar{Y}) \cdot \bar{X} \cdot Y$

$X + \bar{Y} + \bar{X} \cdot Y + 0$

$X + \bar{Y} + \bar{X} \cdot Y$

$X + \bar{Y} + Y \Rightarrow X + 1 = 1$

b. $XYZ + X\bar{Y}Z + \bar{X}YZ + \bar{X}Y\bar{Z}$

$XZ(Y+\bar{Y}) + \bar{X}Y(Z+\bar{Z})$

$XZ + \bar{X}Y \quad \langle \bar{X}Y + XZ \rangle$

c. $\overline{\bar{X}Y} \cdot \overline{X\bar{Z}}$

$\bar{\bar{X}}\bar{Y} + \bar{X}\bar{\bar{Z}} \Rightarrow \bar{X}Y + XZ$

d. $(X+\bar{Y})(\bar{X}+Z)(Y+\bar{Z})$

$(X+\bar{Y})[\bar{X}Y + \bar{X}\bar{Z} + YZ]$

$XYZ + \bar{X}\bar{Y}\bar{Z}$

e. $(W+X+YZ)(\bar{W}+X)(\bar{X}+Y)$

$(W+X+YZ)[\bar{W}\bar{X} + \bar{W}XY + XY] \Rightarrow W\bar{W}\bar{X} + W\bar{W}XY + WXY + \bar{W}\bar{X}XY + \bar{W}XYZ + \bar{X}XYZ$

$XY[\bar{W}\bar{X} + \bar{W}YZ] + XYZ$

$XY + [\bar{W}YZ(\bar{X}+1)]$

$XY(Z+1) + \bar{W}YZ$

$XY + \bar{W}YZ$

f. $WX\bar{Z} + \bar{X}YZ + W\bar{X}\bar{Y} + XYZ + \bar{W}YZ$

$WX(\bar{Y}+\bar{Z}) + \bar{W}YZ + YZ(\bar{X}+X)$

$WX(\bar{Y}+\bar{Z}) + \bar{W}YZ + YZ$

$WX(\bar{Y}+\bar{Z}) + YZ$

$WX(\bar{Y}+\bar{Z}) + (\bar{Y}+\bar{Z})$ } Absorption theorem and De Morgan's theorem

$WX + (\bar{Y}+\bar{Z})$

$WX + YZ$

g. $\bar{W}\cdot X\cdot Z + W\cdot Z + X\cdot Y\cdot\bar{Z} + \bar{W}\cdot X\cdot Y$

$Z(\bar{W}\cdot X + W) + X\cdot Y(\bar{W}+\bar{Z})$

$Z(X+W) + \bar{W}X\cdot Y + X\cdot Y\cdot\bar{Z}$

$X\cdot Z + W\cdot Z + \bar{W}\cdot X\cdot Y + X\cdot Y\cdot\bar{Z}$

$X(Z + Y\cdot\bar{Z}) + W\cdot Z + \bar{W}\cdot X\cdot Y$

$X\cdot Z + X\cdot Y + W\cdot Z + \bar{W}\cdot X\cdot Y$

$X\cdot Z + W\cdot Z + X\cdot Y + \bar{W}\cdot X\cdot Y$

↳ $X\cdot Y + X\cdot Z + W\cdot Z$

k. ✍

i. $(X+Y+Z)(\bar{X}+Y+Z)(\bar{X}+Y+\bar{Z})$

$(X+Y+Z)[\bar{X} + \bar{X}\cdot Y + \bar{X}\cdot Z + \bar{X}\cdot Y + Y + Y\cdot\bar{Z} + \bar{X}\cdot Z + Y\cdot Z]$

$(X+Y+Z)[\bar{X} + \bar{X}\cdot Y + \bar{X}\cdot Z + Y\cdot Z + Y\cdot\bar{Z} + Y]$ // $\langle \bar{X}(1+Y) \quad Y(Z+1)\rangle$

$(X+Y+Z)[\bar{X} + \bar{X}\cdot Z + Y\cdot Z + Y]$

$(X+Y+Z)[\bar{X} + Y]$

$X\cdot Y + \bar{X}\cdot Y + Y + \bar{X}\cdot Z + Y\cdot Z$; $(X+\bar{X})\cdot Y = 1\cdot Y = Y$,

$Y + Y + \bar{X}\cdot Z + Y\cdot Z$

$Y + \bar{X}Z + Y\cdot Z$; $Y(1+Z) = Y$

↳ $\bar{X}\cdot Z + Y$

EXERCISE 2.4 ; Equivalent Logic function

Verify the following equation equalities

g. $A\cdot B\bar{C} + A\cdot C + \bar{B}\cdot C \Rightarrow C \oplus (\bar{A}\cdot B)$

$A\cdot B\bar{C} + C(A+\bar{B})$

$\bar{A}\cdot B\cdot\bar{C} + C(\bar{A}+B)$

$\bar{A}\bar{C}\cdot B\bar{C} + C(\bar{A}\cdot B)$

$\bar{C}(\bar{A}\cdot B) + C(\overline{\bar{A}\cdot B}) \Rightarrow C \oplus (\bar{A}\cdot B)$ ✓

b) $A \oplus B \oplus (A+B) + \bar{A}\cdot\bar{B} = A \odot B \quad \langle AB + \bar{A}\bar{B} \rangle$

$A \oplus [\bar{B}\cdot(A+B) + B\cdot(\overline{A+B})] + \bar{A}\cdot\bar{B}$

$A \oplus [A\bar{B} + B\cdot(\bar{A}\cdot\bar{B})] + \bar{A}\cdot\bar{B}$

$A \oplus [A\bar{B}] + \bar{A}\cdot\bar{B}$

$\bar{A}\cdot(\overline{A\bar{B}}) + A\cdot(\overline{A\bar{B}}) + \bar{A}\cdot\bar{B}$

$0 + A(\bar{A}+B) + \bar{A}\cdot\bar{B}$

↳ $A\cdot B + \bar{A}\cdot\bar{B} \Rightarrow A \odot B$ ✓

c) $(A+B) \oplus (\bar{A}+C) + A = A + B$

$(A+B)(\bar{A}+C) + \overline{(A+B)(\bar{A}+C)} + A$

$AC + \bar{A}B + BC + (\bar{A}\cdot\bar{B})(A\cdot\bar{C}) + A$

$AC + \bar{A}B + BC + A$

$\bar{A}\cdot B + A\cdot C + B\cdot C + A$

$\bar{A}\cdot B + AC + A$

$\bar{A}\cdot B + A(C+1)$

↳ $\cdot A + B_a = A + B$ ✓

d) $(A \odot B) \oplus (A\cdot\bar{B}) + A\cdot\bar{B} = A + B$

$(A\cdot B + \bar{A}\bar{B}) \oplus (A\cdot\bar{B}) + A\cdot\bar{B}$

$\overline{(AB+\bar{A}\bar{B})}(A\cdot\bar{B}) + (AB+\bar{A}\cdot\bar{B})(\overline{A\cdot\bar{B}}) + A\cdot\bar{B}$

$(\bar{A}\bar{B}\cdot\bar{\bar{A}}+\bar{B})(A\cdot\bar{B}) + (A\cdot B + \bar{A}\cdot\bar{B})(\bar{A}+B) + A\cdot\bar{B}$

$(\bar{A}+\bar{B})(A+B)(A\cdot\bar{B}) + (A\cdot B + \bar{A}\cdot\bar{B})(\bar{A}+B) + A\cdot\bar{B}$

$(\bar{A}\bar{B} + A\bar{B})(A\cdot\bar{B}) + A\cdot\bar{A}\cdot\bar{B} + A\cdot B + A\cdot\bar{B}$

$\bar{A}\cdot\bar{B} + \bar{A}\cdot\bar{B} + A\cdot B + A\cdot\bar{B}$

$A\cdot\bar{B} + A\cdot B + \bar{A}\cdot\bar{B}$

$A(\bar{B}+B) + \bar{A}\cdot\bar{B}$

$A + \bar{A}\cdot\bar{B}$

$\hookrightarrow A + \bar{B}$

e $\bar{A}\cdot\bar{B}\bar{C} + A\bar{C}\cdot\bar{D} + \bar{A}\cdot B C\cdot D + A\cdot B\bar{C}\cdot D + A\cdot C\cdot\bar{D} + A\cdot\bar{B}\cdot C = A\cdot C \oplus \bar{A}\cdot\bar{C} \oplus B\cdot D$

$[A\cdot B\bar{C}D + \bar{A}\cdot B\cdot C\cdot D] + [\bar{A}\cdot\bar{B}\cdot\bar{C} + A\cdot\bar{B}\cdot C] + [\bar{A}\cdot\bar{C}\cdot\bar{D} + A\cdot C\cdot\bar{D}]$

$(A\bar{C} + \bar{A}C)(B\cdot D) + \bar{B}(\bar{A}\bar{C} + A\cdot C) + \bar{D}(\bar{A}\bar{C} + AC)$

$(A+C)(\bar{A}+\bar{C})(B\cdot D) + (\bar{A}\cdot\bar{C} + A\cdot C)(\bar{B}+\bar{D})$

$[\overline{(\bar{A}\cdot\bar{C})(A\cdot C)}](B\cdot D) + (\bar{A}\cdot\bar{C} + A\cdot C)(\overline{B\cdot D})$

$(\overline{A\bar{C} + A\cdot C})(B\cdot D) + (\bar{A}\cdot\bar{C} + A\cdot C)(\overline{B\cdot D})$

$\bar{A}\cdot\bar{C} + AC \oplus B\cdot D$

Rewritten as

$\bar{A}\cdot\bar{C} + \bar{A}\cdot\bar{C} + AC + A\cdot C$

$(A+C)(A\cdot C) = A\cdot C + A C$

$\therefore (\bar{A}+\bar{C})(\bar{A}\cdot\bar{C}) = \bar{A}\bar{C} + \bar{A}\cdot\bar{C}$

$\hookrightarrow (\bar{A}+C)(\bar{A}-\bar{C}) + (A+C)(A\cdot C) \oplus B\cdot D$

$(\overline{A\cdot C})(\overline{A\cdot C}) + (A C)(\overline{A\cdot C}) \oplus BD$

$\hookrightarrow A\cdot C \oplus \bar{A}\cdot\bar{C} \oplus B\cdot D$ ✓

f $A\cdot B\bar{D} + \bar{B}\cdot C\cdot D + A\cdot B\cdot\bar{C} + BC\cdot D + \bar{A}\cdot C\cdot D = A\cdot B + C\cdot D$

$C\cdot D(B+\bar{B}) + A B\bar{D} + A B C + \bar{A}\cdot C\cdot D$

$C\cdot D + A\cdot B\bar{D} + A\cdot B\bar{C} + \bar{A}\cdot C\cdot D$

$C\cdot D(1+\bar{A}) + A\cdot B\cdot\bar{D} + A\cdot B\bar{C}$

$C\cdot D + A\cdot B(\bar{C}+\bar{D})$

$C\cdot D + A\cdot B(\bar{C}+\bar{D})$

$\hookrightarrow \overline{\bar{C}+\bar{D}} + A\cdot B(\bar{C}+\bar{D})$

$A\cdot B + \overline{\bar{C}+\bar{D}}$

$\hookrightarrow A\cdot B + C\cdot D$.

EXERCISE 25;

Simplification Of the Functions Using The Karnaugh Map Method

Simplify the following with K-Map

$F(A,B,C,D) = \Sigma m(0,2,3,4,5,8,11,13,13,14,15)$

AB \ CD	00	01	11	10
00	1		1	1
01	1	1		
11	1	1	1	1
10	1		1	

The (4) pairs; $\bar{C}\cdot\bar{D} + A\cdot B + B\cdot\bar{C}$

Pairs of two; $\bar{A}\cdot\bar{B}\cdot C + \bar{B}\cdot C\cdot D$

$\hookrightarrow A\cdot B + B\cdot\bar{C} + \bar{C}\cdot\bar{D} + \bar{A}\cdot\bar{B}\cdot C + \bar{B}\,C\cdot D$

AB \ CD	00		11	10
00	1		1	1
01	1	1		
11	1	1	1	1
10	1		1	

4 pairs; $\bar{C}\cdot\bar{D} + A\cdot B + B\cdot\bar{C}$

Pairs of two; $\bar{A}\cdot\bar{B}\cdot C + A\cdot C\cdot D$

$\hookrightarrow A\cdot B + B\cdot\bar{C} + \bar{C}\cdot\bar{D} + \bar{A}\cdot\bar{B}\cdot C + A\cdot C\cdot D$

Q4 $F(A,B,C,D) = \sum m(0,1,3,4,6,7,8,9,11,12,13,14,15)$

(i)

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	1	1
11	1	1	1	1
10	1	1	1	0

Pair of 4s; $\bar{C}\bar{D} + A\bar{D} + \bar{B}\bar{D} + B\cdot C$

(ii)

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	1	1
11	1	1	1	1
10	1	1	1	0

Pair of 4s, $C\cdot D + A\cdot B + B\cdot\bar{D} + \bar{B}\cdot\bar{C}$

$G(A,B,C,D) = \sum m(0,2,4,5,10,12,15) + \sum x(8,14)$ ← Do not care

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	0	0
11	1	0	1	x
10	x	0	0	1

Pair of 4s; $\bar{C}\bar{D}$

Pair of 2s; $\bar{A}\cdot B\cdot\bar{C} + A\cdot B\cdot C + \bar{B}\cdot C\cdot\bar{D}$

↳ $\bar{A}\cdot B\bar{C} + A\cdot B\cdot C + \bar{C}\cdot\bar{D} + \bar{B}\cdot C\cdot\bar{D}$

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	0	0
11	1	0	1	x
10	x	0	0	1

could go diagonally

Pair of 4s; $\bar{C}\cdot\bar{D} + \bar{B}\cdot\bar{D}$

Pair of 2s; $\bar{A}\cdot B\cdot\bar{C} + A\cdot B\cdot\bar{C}$

↳ $\bar{A}\cdot B\cdot\bar{C} + A\cdot B\cdot C + \bar{B}\cdot\bar{D} + \bar{C}\cdot\bar{D}$

$F(A,B,C,D) = \sum_m(1,3,6,8,11,14) + \sum_x(2,5,12,13,15)$

Could plot it straight or write out the logic ~~equation~~ expression with each cell in the Karnaugh map equaling the numbers given no need for an expression just plotting.

AB \ CD	00	01	11	10
00	0	1	1	X
01	0	X	0	1
11	X	X	X	1
10	1	0	1	0

$\Rightarrow A\bar{C}\bar{D} + B\bar{C}\bar{D} + \bar{A}\bar{B}D + A\bar{C}D$,

EXERCISE 2.6; Analyses of the Circuit with NAND/NOR gates

Fig 2-63(a) logic expression;

First gate; $\overline{A+B}$, $\overline{A+C}$

Second gate; $\overline{\overline{A+B}+A}$, $\overline{\overline{A+C}+A}$

Third gate; $\overline{\overline{\overline{A+B}+A} + \overline{\overline{A+C}+A}}$

$F = \overline{\overline{\overline{A+B}+A} + \overline{\overline{A+C}+A}}$

$(\overline{A+B+A}) \cdot (\overline{A+C+A})$

$(\bar{A} \cdot \bar{B} + A)(\bar{A} \cdot \bar{C} + A)$

$(A + B)(A + \bar{C})$

$A + A\bar{C} + A\bar{B} + \bar{B} \cdot \bar{C}$

$A(1 + \bar{B} + \bar{C}) + \bar{B} \cdot \bar{C}$

$A(1 + \bar{C}) + \bar{B} \cdot \bar{C}$

$F = A + \bar{B} \cdot \bar{C}$

Fig 2.63(b) Logic expression;

first gate ⇒ $\overline{\bar{A} \cdot B}$ $\overline{A \cdot \bar{C}}$

Second gate ⇒ $\overline{\overline{\bar{A} \cdot B} \cdot \bar{A}}$ $\overline{\overline{A \cdot \bar{C}} \cdot \bar{A}}$

Last gate ⇒

↳ $\overline{\overline{\bar{A} \cdot B \cdot \bar{A}} \cdot \overline{\bar{A} \cdot \bar{C} \cdot \bar{A}}}$

↳ $\overline{\bar{A} \cdot B \cdot \bar{A}} + \overline{\bar{A} \cdot \bar{C} \cdot \bar{A}}$

$(A + B \cdot \bar{A}) + (A + C \cdot \bar{A})$

$G \Rightarrow \bar{A} \cdot B + \bar{A} \cdot C$

↳ $\bar{A}(B + C)$

Verify $F = G$

$\therefore \bar{G} = \overline{\bar{A}(B + C)}$

$\bar{G} = A + \overline{(B + C)}$

$\bar{G} = A + (\bar{B} \cdot \bar{C}) = F$

$\therefore F = G$

EXERCISE 2.7 ; Canonical Form Of A Logic Function

Figure 2-6a(a)

1st GATE ↳ $\overline{A+C}$, $\overline{B+C}$, $\overline{B+D}$

2ND GATE ↳ $\overline{\overline{A+C}+B}$, $\overline{A+\overline{B+C}}$, $\overline{\overline{B+C}+D}$, $\overline{C+\overline{B+D}}$

3RD GATE ↳ $\overline{\overline{A+C}+B}+\overline{A+\overline{B+C}}+\overline{\overline{B+C}+D}+\overline{C+\overline{B+D}}$ ⇒ F logic expressions

Figure 2-6(b)

1st GATE ↳ $\overline{A \cdot C}$, $\overline{B \cdot C}$, $\overline{B D}$

2ND GATE ↳ $\overline{\overline{A \cdot C} \cdot B}$, $\overline{A \cdot \overline{B \cdot C}}$, $\overline{\overline{B C} \cdot D}$, $\overline{\overline{C} \cdot \overline{B \cdot D}}$

3RD GATE (G) ⇒ $\overline{\overline{A C} \cdot B} \cdot \overline{A \cdot \overline{B C}} \cdot \overline{\overline{B C} \cdot D} \cdot \overline{\overline{C} \cdot \overline{B \cdot D}}$ ⇒ G logic expression

F Expressed as Sum of products;

↳ $(\overline{\overline{A+C}+B}) \cdot (A+\overline{B+C}) \cdot (\overline{B+C}+D) \cdot (C+\overline{B+D})$

$(\overline{A} \cdot \overline{C} + B)(A+\overline{B} \cdot \overline{C}) \cdot (\overline{B} \cdot \overline{C}+D) \cdot (C+\overline{B} \cdot \overline{D})$

$(\overline{A} \cdot \overline{B} \cdot \overline{C} + A B) \cdot (\overline{B} \cdot C \cdot \overline{D} + C \cdot D)$

$\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot C \cdot D$

$\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$ ⇒ 0000 ⇒ 0

$A \cdot B \cdot C \cdot D = 1111$ ⇒ 15

Canonical Form ⇒ $\Sigma M(0,15)$

G Expressed as Product-of-Sum;

↳ ~~$(\overline{A} \cdot C)(B) + (A \cdot \overline{B} \cdot C)(A)(C \cdot D)) + (C \cdot \overline{B} \cdot D)$~~

↳ $(A \cdot C + \overline{B})(\overline{A} + B \cdot C)(B \cdot C + \overline{D})(\overline{C} + B \cdot D)$

↳ $(\overline{A} \cdot \overline{B} + A \cdot B \cdot C)(B \cdot D + \overline{C} \cdot \overline{D})$

$\bar{A}\cdot\bar{B}\cdot\bar{C}\cdot\bar{D} + A\cdot B\cdot C\cdot D$

$(\bar{A}\cdot\bar{B}\cdot\bar{C}\cdot\bar{D})(A\cdot B\cdot C\cdot D)$

$(A+B+C+D)(\bar{A}+\bar{B}+\bar{C}+\bar{D})$

Canonical = $\Pi m(0,15)$

EXERCISE 2.8 → Simplification of Logic Circuits

~~Ex 2.8~~ Fig 2.65(a),

Logical expression;

1st Logic: $A\cdot C$, $\overline{A\cdot C}$, $A+B+C$

2nd Gate: $(\overline{A\cdot C})(A+B+C)$

3rd Gate: $A\cdot C + (\overline{A\cdot C})(A+B+C)$ ← Can use absorption theorem here

$A\cdot C + (\bar{A}+\bar{C})(A+B+C)$

~~$A\cdot C + \bar{A}B + \bar{A}\cdot C + A\cdot\bar{C} + B\cdot\bar{C}$~~ MISTAKE;

$\bar{A}\cdot B + AC +$

$A\cdot C + (A+B+C)$

$A(C+1) + B + C$

$G = A+B+C$

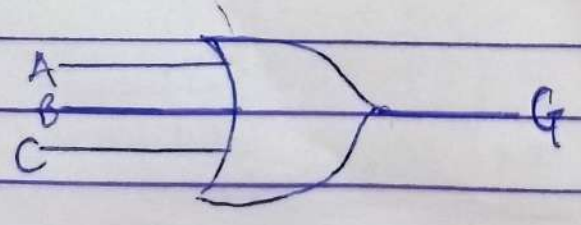


Fig 2.65 (b) Logical Expression;

1st Gate; $A \oplus B$, $A \oplus \bar{C}$

2nd Gate $[A \oplus B] \oplus [A \oplus C]$

$$H = (A \oplus B) \oplus (A \oplus \bar{C})$$

$$H \Rightarrow (\bar{A} \cdot B + A \cdot \bar{B}) \oplus (\bar{A} \cdot \bar{C} + A \cdot C)$$

$$H \Rightarrow \overline{(\bar{A}B + A\bar{B})}(\bar{A}\bar{C} + AC) + (\bar{A} \cdot B + A\bar{B})\overline{(\bar{A}\bar{C} + AC)}$$

$$(\bar{A}B \cdot A\bar{B})(A\bar{C} + AC) + (\bar{A}B + A\bar{B})(\bar{A}\bar{C} \cdot \bar{A}C)$$

$$[(A+B)(\bar{A}+\bar{B})][\bar{C}\bar{A}\cdot\bar{C} + AC] + [(\bar{A}\cdot B + A\cdot\bar{B})(A+C)(\bar{A}+\bar{C})]$$

$$[A\cdot B + \bar{A}\cdot\bar{B}][\bar{A}\bar{C} + AC] + [\bar{A}\cdot B + A\cdot\bar{B}][A\cdot\bar{C} + \bar{A}C]$$

$$[A\cdot B\cdot C + \bar{A}\bar{B}\bar{C}] + [\bar{A}\cdot B\cdot C + A\cdot\bar{B}\bar{C}] \;;\; BC(A+\bar{A}) + \bar{B}\bar{C}(A+\bar{A})$$

$$B\cdot C + \bar{B}\cdot\bar{C}$$

$$H \hookrightarrow B \odot C = \overline{B \oplus C}$$

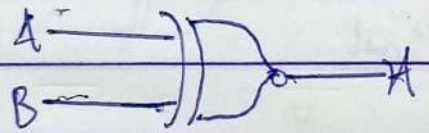


Fig 2.65 (c) Logic Expression;

1st Gate; $B + 0 = B$, $\bar{C}$, $C \cdot D$

$\hookrightarrow \bar{B}$

2nd Gate; $\overline{\bar{A} \cdot \bar{B}}$, $\bar{B} \cdot \bar{C}$, $\bar{C} + C \cdot D + D \Rightarrow \bar{C} + D$

3rd Gate; $R \Rightarrow \overline{\bar{A} \cdot \bar{B}} \cdot \bar{B} \cdot \bar{C} \cdot (\bar{C} + D)$

$$(\bar{A} + B)\bar{B} \cdot \bar{C}(\bar{C} + D)$$

$$\bar{A} \cdot \bar{B} \cdot \bar{C}(\bar{C} + D)$$

$$\bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot D$$

$$R = \bar{A} \cdot \bar{B} \cdot \bar{C}(1 + D)$$

$$R = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

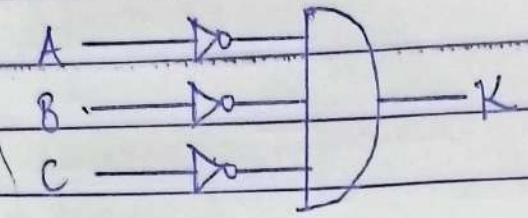


Fig 2-65 (d) Logic Expression;

$$K = \overline{(A \cdot \bar{B}) + \bar{C}} + \overline{A \oplus \bar{B}} + \overline{A + \bar{C}}$$

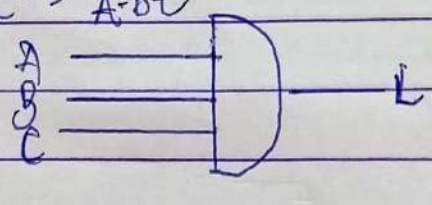
$$(\bar{A} \cdot B + \bar{C}) \cdot (A \oplus \bar{B}) + (A + \bar{C})$$

$$(\bar{A} + B \cdot C)(A \oplus B) \cdot (A + \bar{C})$$

$$(\bar{A} \cdot C + B \cdot C)(A \cdot \bar{B} + A \cdot B)(A + \bar{C})$$

$$(\bar{A} \cdot \bar{B} \cdot C + A \cdot B \cdot C)(A + \bar{C})$$

$K \Rightarrow A \cdot B \cdot C$



Logic Circuit

Fig 2-65 (e) Logic Expression;

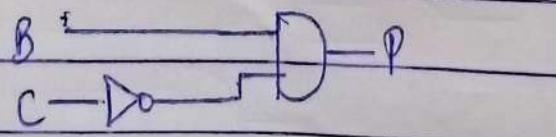
$$P = A \cdot B \cdot \overline{\bar{B} \cdot C} + A \cdot D \cdot \overline{\bar{B} + C} + \overline{\bar{B} + C}$$

$$P = A \cdot B \cdot \bar{B} + \bar{C} + A \cdot D \cdot B \cdot \bar{C} + B \cdot \bar{C}$$

$$P = A \cdot B \cdot \bar{C} + A \cdot D (B \cdot \bar{C}) + B \cdot \bar{C}$$

$$P = [A + A \cdot D + 1] B \cdot \bar{C}$$

$$P = B \cdot \bar{C}$$



Logic Circuit

Fig 2.65 (f) Logic Expressions:

1st Gate ⇒ $B \oplus C$, $\overline{A \oplus B}$, $A \oplus B$, $\overline{B \oplus C}$

2nd Gate ⇒ $(\overline{A \oplus B})(B \oplus C)$, $(A \oplus B)(\overline{B \oplus C})$

3rd Gate Q ⇒ $(\overline{A \oplus B})(B \oplus C) \oplus (A \oplus B)(\overline{B \oplus C})$

This equal

↳ $(A \oplus B) \oplus (B \oplus C)$

Factorization theorem ⇒ $(A \oplus C) \oplus (B \oplus B)$

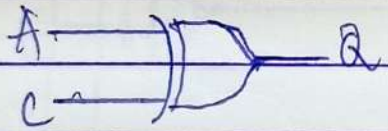
The NOR of a single gate is $= 0$

$B \oplus B = \bar{B}B + B\bar{B}$

↳ 0

$\therefore (A \oplus C) \oplus 0$

Q ⇒ $A \oplus C$



EXERCISE 2.9 ⇒ Implementation of the function H

$H(A,B,C) \Rightarrow A\cdot\bar{B} + \bar{A}\cdot B + B\cdot\bar{C}$

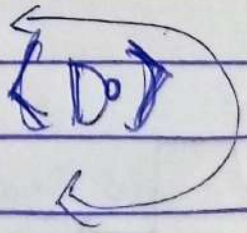
↳ $A\cdot\bar{B} + (\bar{A}+\bar{C})B$

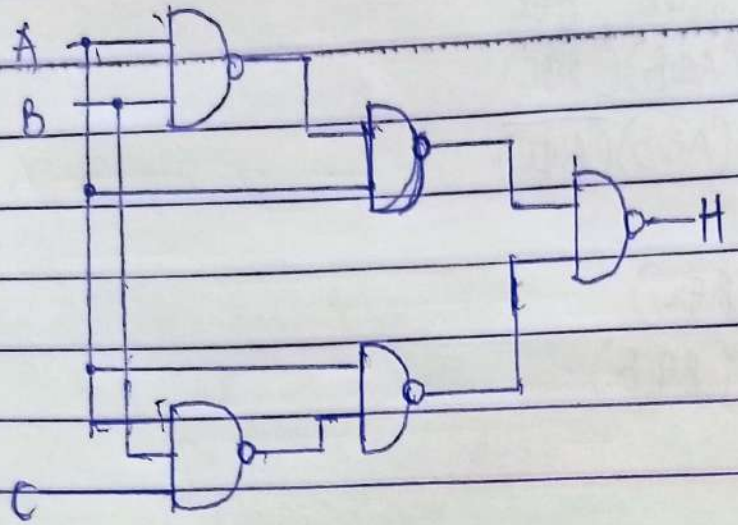
$A(\bar{A}+\bar{B}) + (\bar{A}+\bar{C})B$

$A\cdot\overline{A\cdot B} + \overline{A\cdot C}\cdot B$

Implementing only use a two input NAND

So $\overline{A\cdot\overline{A\cdot B}}\cdot\overline{\overline{A\cdot C}\cdot B}$





Logic Circuit for the two input NAND GATES

EXERCISE 2-10 ⇒ Implementation of the function Y.

$$Y = (A \cdot B) \oplus (\bar{B} \cdot \bar{C})$$

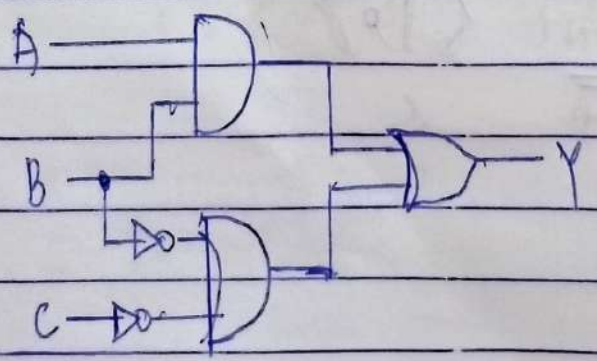
$$Y = (A \cdot B)\overline{(\bar{B} \cdot \bar{C})} + \overline{(A \cdot B)}(\bar{B} \cdot \bar{C})$$

$$Y = (\bar{A} + \bar{B})(\bar{B} \cdot \bar{C}) + (A \cdot B)(B + C)$$

$$Y = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{B} \cdot \bar{C} + A \cdot B + A \cdot B \cdot C$$

$$Y = \bar{B} \cdot \bar{C}(\bar{A} + 1) + A \cdot B(1 + C)$$

$$Y = \bar{B} \cdot \bar{C} + A \cdot B$$



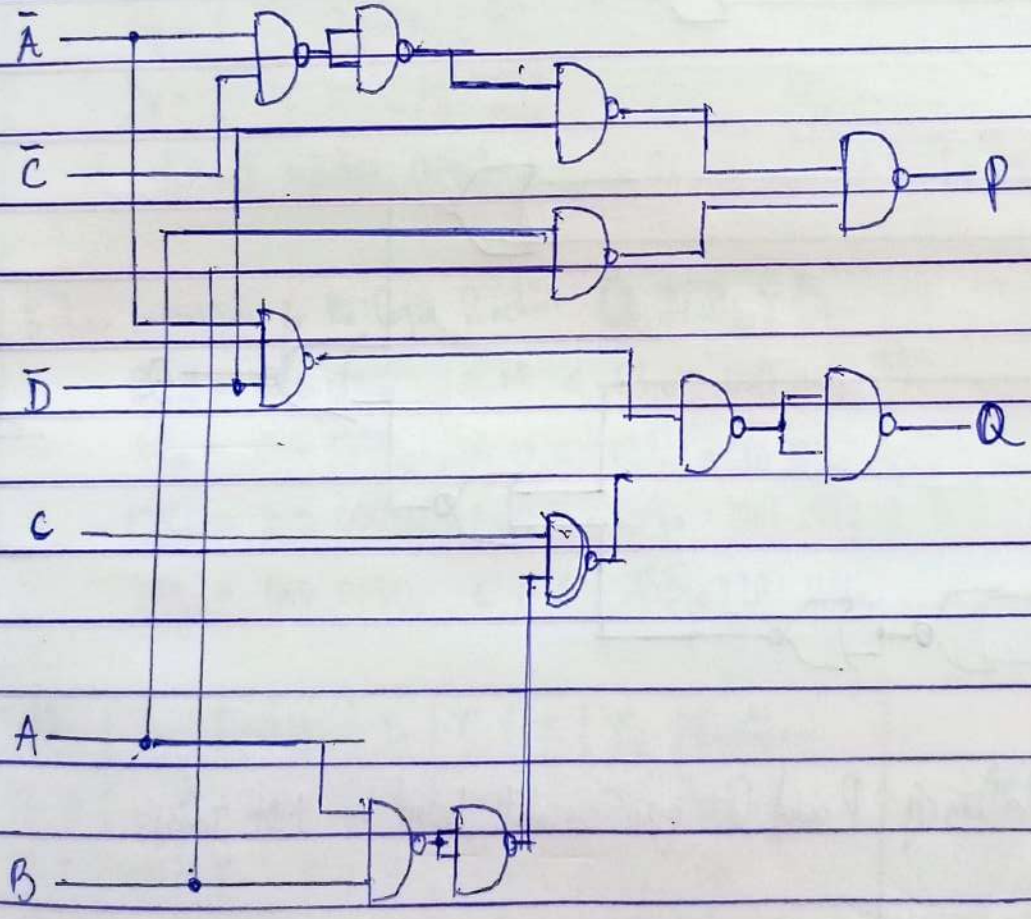
EXERCISE 2-11 ⇒ Implementation of Circuits with NAND/OR Gates.

$P(A,B,C,D) = A \cdot B + \bar{A} \cdot \bar{C} \cdot \bar{D}$

$Q(A,B,C,D) = (\bar{A}+\bar{B}+\bar{C})(A+D)$

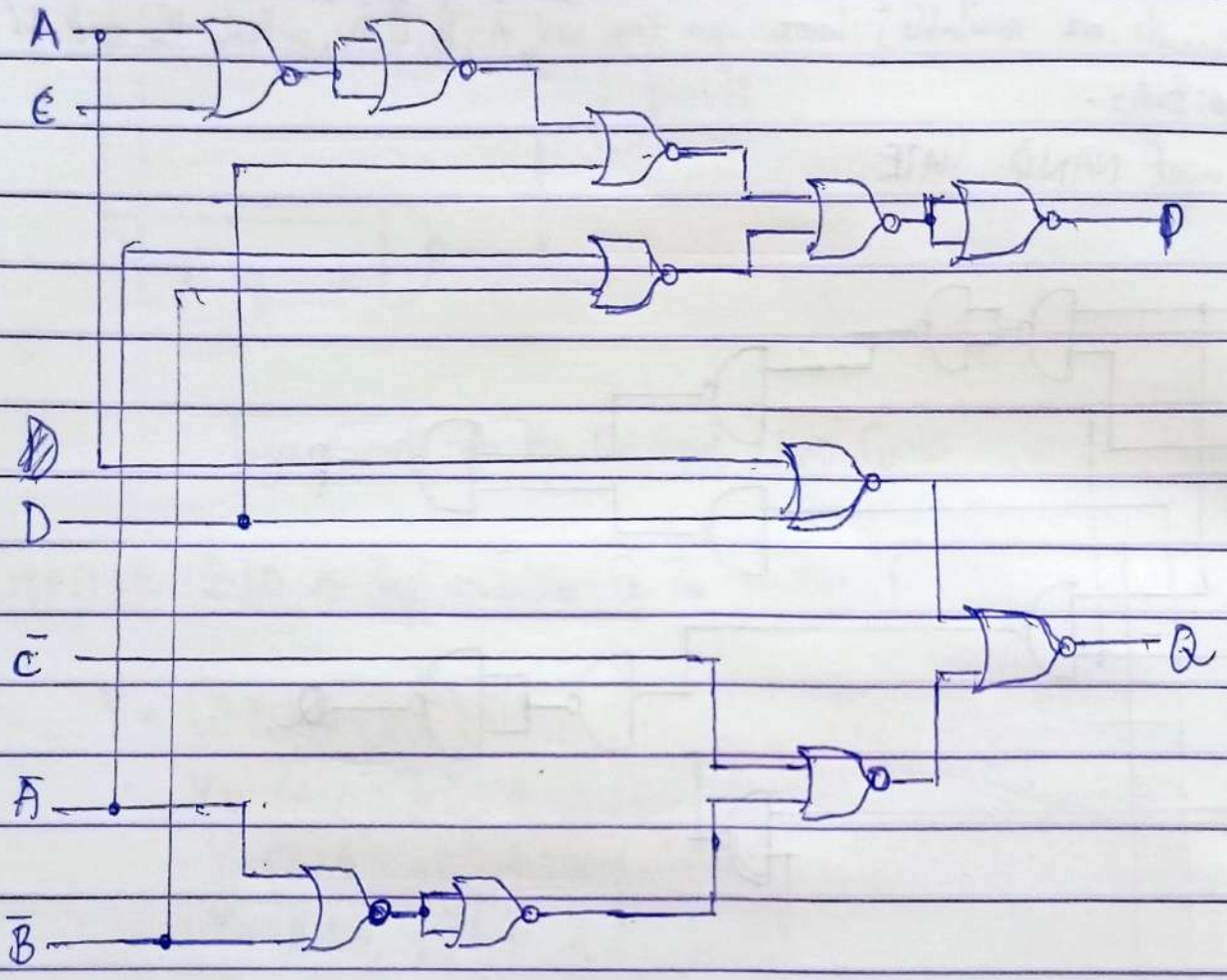
Complements are available; hence you can use $\bar{A}, \bar{B}, \bar{C}, \bar{D}$ directly without the need for an inverter Buffer.

2 input NAND GATE



Implementation of P and Q logic circuit based on NAND Gates.

2 input NOR Gate



Implementation of P and Q logic circuit based on NOR gates

EXERCISE 2.12 – Control Circuit For A Switcher.

Switch in four directions packages identified by an eight-bit binary code:
$I_7 I_6 I_5 I_4 I_3 I_2 I_1 I_0$

I_7 ⇒ Most significant bit

$D_0 \Rightarrow 1$ (active)

$D_1 = 1$ if $32 \leq N \leq 63$

$D_2 = 1$ if $64 \leq N \leq 127$

$D_3 = 1$ if $128 \leq N \leq 159$

$D_4 = 1$ if $192 \leq N \leq 255$

N ; decimal number corresponding to the code $I_7 I_6 I_5 I_4 I_3 I_2 I_1 I_0$

a) Boolean expression for the logic function (D_1, D_2, D_3 & D_4)

$32_{10} = 0010\ 0000_2 \leq N \leq 63_{10} = 0011\ 1111_2$

$64_{10} = 0100\ 0000_2 \leq N \leq 127_{10} = 0111\ 1111_2$

$128_{10} = 1000\ 0000_2 \leq N \leq 159_{10} = 1001\ 1111_2$

$192_{10} = 1100\ 0000_2 \leq N \leq 255_{10} = 1111\ 1111_2$

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Functions	
0	0	1	0	0	0	0	0	32 D_1	$D_1 = \bar{I}_7 \cdot \bar{I}_6 \cdot I_5$
0	0	1	1	1	1	1	1	63	
0	1	0	0	0	0	0	0	64 D_2	$D_2 = \bar{I}_7 \cdot I_6$
0	1	1	1	1	1	1	1	127	
1	0	0	0	0	0	0	0	128 D_3	$D_3 = I_7 \cdot \bar{I}_6 \cdot \bar{I}_5$
1	0	0	1	1	1	1	1	159	
1	1	0	0	0	0	0	0	192 D_4	$D_4 = I_7 \cdot I_6$
1	1	1	1	1	1	1	1	255	

$D_0 = 1$; Meaning its active

$D_4 = I_7 \cdot I_6$

$D_3 = I_7 \bar{I}_6 \cdot I_5$

$D_2 = \bar{I}_7 \cdot I_6$

$D_1 = \bar{I}_7 \cdot \bar{I}_6 I_5$

$\therefore \overline{D_0} = D_4 + D_3 + D_2 + D_1$

$\hookrightarrow D_0 = \overline{D_4 + D_3 + D_2 + D_1}$

$D_0 \Rightarrow$

$\hookrightarrow \overline{I_7 \cdot I_6 + I_7 \bar{I}_6 \cdot \bar{I}_5 + \bar{I}_7 I_6 + \bar{I}_7 \cdot \bar{I}_6 \cdot I_5}$

$\hookrightarrow \overline{I_7 (I_6 + \bar{I}_6 \cdot \bar{I}_5) + \bar{I}_7 (I_6 + \bar{I}_6 \cdot I_5)}$

$\hookrightarrow \overline{I_7 (I_6 + \bar{I}_5) + \bar{I}_7 (I_6 + I_5)}$

$\hookrightarrow \overline{I_7 \cdot I_6 + I_7 \cdot \bar{I}_5 + \bar{I}_7 I_6 + \bar{I}_7 \cdot I_5}$

$\hookrightarrow \overline{I_6 (\bar{I}_7 + I_7) + \bar{I}_7 \cdot I_5 + I_7 \cdot \bar{I}_5}$

$\hookrightarrow \overline{I_6 + \bar{I}_7 \cdot I_5 + I_7 \cdot \bar{I}_5}$

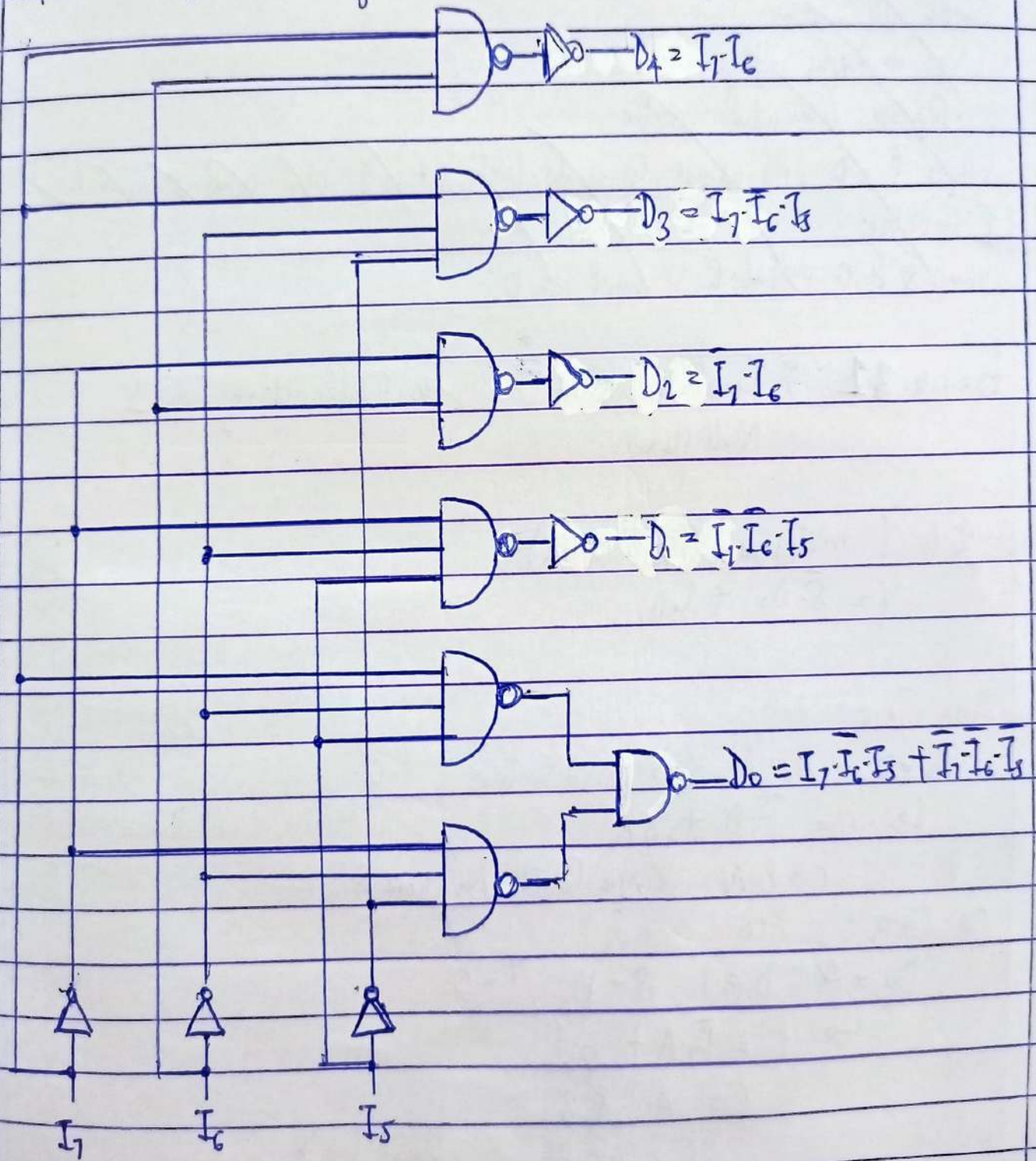
$\hookrightarrow \bar{I}_6 (\bar{I}_7 \cdot \bar{I}_5)(I_7 \cdot \bar{I}_5)$

$\hookrightarrow \bar{I}_6 (I_7 + \bar{I}_5)(\bar{I}_7 + I_5)$

$\hookrightarrow \bar{I}_6 [I_7 \cdot I_5 + \bar{I}_7 \cdot \bar{I}_5]$

$D_0 \Rightarrow I_7 \bar{I}_6 \cdot I_5 + \bar{I}_7 \cdot \bar{I}_6 \cdot \bar{I}_5$

b Implement the functions using only inverters and NAND gates ; inputs, three inputs.



Logic Circuit

CHAPTER 3

Exercise 1-1

Switch Debouncing

V_{cc} ⇒ Supply voltage

R_p ⇒ Polarization resistor

When $\bar{R} = 0$, the output Q of the latch set to 1 when signal $\bar{S}$ activates at 1 initially.

When $\bar{S}$ is 0, output Q is reset to 0

Exercise 1 ; Implementation Of Logic Gates Using 2:1 Multiplexer.

2 to 1 multiplexer (2:1 MUX)

$Y = \bar{S} \cdot D_0 + S \cdot D_1$

AND GATE; A·B

$D_0 = 0$ $D_1 = A$ $S = B$ ⟨Activator⟩ $Y = C$

↳ $C = \bar{B} \cdot 0 + B \cdot A$

$C = B \cdot A$ ⟨AND GATE NOTATION⟩

OR GATE; A+B

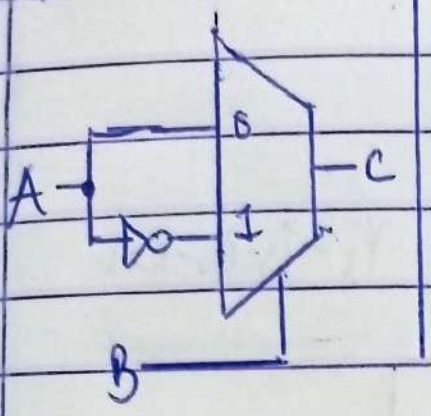
$D_0 = A$, $D_1 = 1$ $S = B$ $Y = C$

↳ $C = \bar{B} \cdot A + B \cdot 1$

$C = A \cdot \bar{B} + B$ ↰

$C = A + B$ ⟨Absorption Law⟩

XOR GATE: ⇒ $A \oplus B$

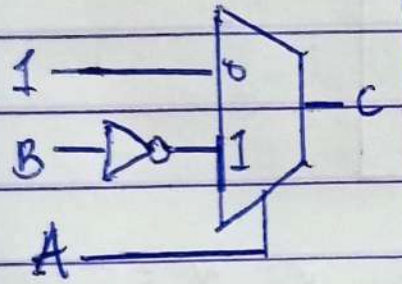


$B = S \quad D_0 = A \quad D_1 = \bar{A} \quad C = Y$

$C = \bar{B} \cdot A + B \cdot \bar{A}$

$C = A \oplus B$

NAND GATE: ⇒ $\overline{A \cdot B}$



$A = S \quad D_0 = 1 \quad D_1 = \bar{B} \quad C = Y$

$C = \bar{A} \cdot 1 + A \cdot \bar{B}$

$C = \bar{A} + A\bar{B}$

$C = \bar{A} + \bar{B} \Rightarrow \overline{AB}$

Exercise 3.2: Implementation of 2-Out-of-4 Decoder Using 1:2 MUX

(1) Fig 3.61(a) and Fig 3.61(b)

$Y_0 = \bar{S} \cdot D \quad Y_1 = S \cdot D$

which is a 1-2 demultiplexer

(2) Fig 3.62(a)

Y_0 of first decoder

$\Rightarrow D_1 \cdot EN$

To the Y_0 & Y_1 decoder; $\bar{D}_1 \cdot EN$ is an input D_0 is also an input

for $Y_0 \Rightarrow \bar{D}_1 \cdot D_0 \cdot EN$

for $Y_1 \Rightarrow \bar{D}_1 \cdot D_0 \cdot EN$

→ In the Y_2 & Y_3 decoder; $\overline{EN}\cdot D_1$ is the input and D_0 does

so $Y_2 \Rightarrow D_1 \cdot \overline{D_0} \cdot EN$

$Y_3 \Rightarrow D_1 \cdot D_0 \cdot EN$

$Y_0 = \overline{D_1} \cdot \overline{D_0} \cdot EN$, $Y_1 = \overline{D_1} \cdot D_0 \cdot EN$, $Y_2 = D_1 \cdot \overline{D_0} \cdot EN$ $Y_3 = D_1 \cdot D_0 \cdot EN$.

D_1	D_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	EN
0	1	0	0	EN	0
1	0	0	EN	0	0
1	1	EN	0	0	0

Truth Table;

$Y_3 = D_1 \cdot D_0 \cdot EN$	$Y_2 = D_1 \cdot \overline{D_0} \cdot EN$	$Y_1 = \overline{D_1} \cdot D_0 \cdot EN$	$Y_0 = \overline{D_1} \cdot \overline{D_0} \cdot \overline{EN}$
$D_1=0$ $D_0=0$	$D_1=0$ $D_0=0$	$D_1=0$ $D_0=0$	$D_1=0$ $D_0=0$
$Y_3 = 0\cdot 0\cdot EN = 0$	$Y_2 = 0\cdot \overline{0}\cdot \overline{EN} = 0$	$Y_1 = \overline{0}\cdot 0\cdot EN = 0$	$Y_0 = 1\cdot 1\cdot \overline{EN} = \overline{EN}$
$D_1=0$ $D_0=1$	$D_1=0$ $D_0=1$	$D_1=0$ $D_0=1$	$D_1=0$ $D_0=1$
$Y_3 = 0\cdot 1\cdot EN = 0$	$Y_2 = 0\cdot 0\cdot EN = 0$	$Y_1 = 1\cdot 1\cdot EN = \overline{EN}$	$Y_0 = 1\cdot 0\cdot EN = 0$
$D_1=1$ $D_0=0$	$D_1=1$ $D_0=0$	$D_1=1$ $D_0=0$	$D_1=1$ $D_0=0$
$Y_3 = 1\cdot 0\cdot EN = 0$	$Y_2 = 1\cdot 1\cdot EN = \overline{EN}$	$Y_1 = 0\cdot 0\cdot EN = 0$	$Y_0 = 0\cdot 1\cdot EN = 0$
$D_1=1$ $D_0=1$	$D_1=1$ $D_0=1$	$D_1=1$ $D_0=1$	$D_1=1$ $D_0=1$
$Y_2 = 1\cdot 1\cdot EN = EN$	$Y_2 = 1\cdot 0\cdot \overline{EN} = 0$	$Y_1 = 0\cdot 1\cdot EN = 0$	$P_0 = 0\cdot 0\cdot EN = 0$

Exercise 33; Realization of 2-to-4 decoder using NAND Gates

First Gate $\Rightarrow \overline{D_0 \cdot D_1} \Rightarrow \bar{D}_0 + \bar{D}_1$ also $\Rightarrow \bar{Y}_3$

Second Gate (a) $\Rightarrow \overline{D_0 \cdot \overline{D_0 \cdot D_1}} \Rightarrow \bar{D}_0 + D_0 \cdot D_1$

$\Rightarrow D_1 + D_0 = \bar{Y}_1$

Second Gate (b) $\Rightarrow \overline{D_1 \cdot \overline{D_0 \cdot D_1}} = \bar{D}_1 + D_0 \cdot D_1$

$\Rightarrow \bar{D}_1 + D_0 = \bar{Y}_2$

Third Gate $= \bar{Y}$

$\bar{Y}_0 = \bar{Y}_1 \cdot \bar{Y}_2 \cdot \bar{Y}_3$

$\bar{Y}_0 = (D_1 + \bar{D}_0)(\bar{D}_1 + D_0)(\bar{D}_1 + \bar{D}_0)$

$Y_0 \Rightarrow \overline{(D_1 + \bar{D}_0)} + \overline{(\bar{D}_1 + D_0)} + \overline{(\bar{D}_1 + \bar{D}_0)}$

$Y_0 \Rightarrow (\bar{D}_1 \cdot D_0) + (D_1 \cdot \bar{D}_0) + (D_1 \cdot D_0)$

$Y_0 \Rightarrow D_1 [D_0 + \bar{D}_0] + \bar{D}_1 \cdot D_0$

$Y_0 \Rightarrow D_1 + \bar{D}_1 \cdot D_0$

$Y_0 \Rightarrow D_1 + D_0$

In NAND form;

$Y_0 \Rightarrow \overline{\bar{D}_1 \cdot \bar{D}_0}$ $Y_1 \Rightarrow \overline{\bar{D}_1 \cdot D_0}$ $Y_2 \Rightarrow \overline{D_1 \cdot \bar{D}_0}$ $Y_3 \Rightarrow \overline{D_1 \cdot D_0}$

D_1	D_0	$\bar{Y}_3$	$\bar{Y}_2$	$\bar{Y}_1$	$\bar{Y}_0$
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1

Exercise 34 ⇒ 8:3 Priority Encoder

2^n :- 8:3 :how 3 variables.

Priority encoder needs only one active input (ie at 1).

Logic equation;

We have A, B, C, D, W, Y_0, Y_1, Y_2

First gate (1) ⇒ $D_0 + D_1$

(2) ⇒ $D_2 + D_3$

(3) ⇒ $D_4 + D_5$

(4) ⇒ $D_6 + D_7$

Output of First gate (2) =

$\therefore B = D_2 + D_3$

Output of First gate (4) = D

$\therefore D = D_6 + D_7$

First Schematic Diagram (1) ⇒

Input ⇒ $\bar{B} \cdot D_1 + B \cdot D_3$

$\therefore D_1(\overline{D_2 + D_3}) + D_3(D_2 + D_3)$ which is equal to A

Diagram (2);

Input ⇒ $\bar{D} \cdot D_5 + D \cdot D_7$

$\Rightarrow D_5(\overline{D_6 + D_7}) + D_7(D_6 + D_7)$

Whose output is equal to C

Second Gates (1) ⇒ $D_0 + D_1 + B$

⇒ $D_0 + D_1 + D_2 + D_3$; which we will call X

(2) ⇒ $D_4 + D_5 + D$

$D_4 + D_5 + D_6 + D_7$; which is equal to Y_2

Re Arranging and Re-Drawing with the values gotten.

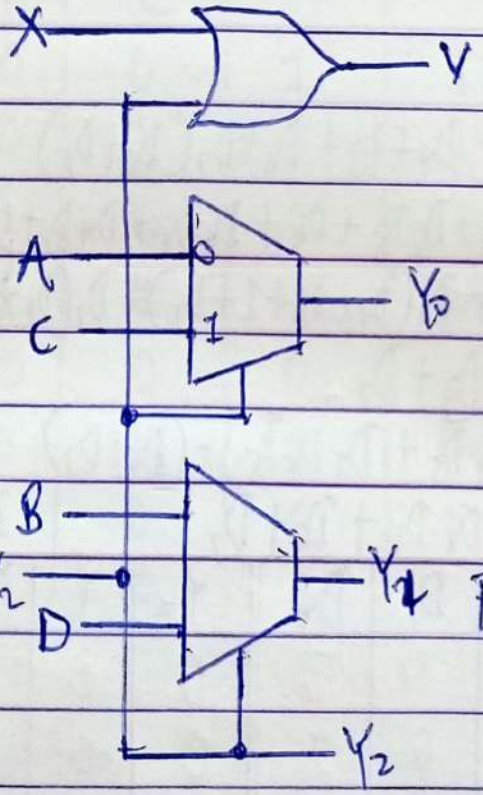


Fig: Shorter Circuit of the 8:3 Priority Encoder

$\therefore V \Rightarrow X + Y_2 \Rightarrow D_0 + D_1 + D_2 + D_3 + D_4 + D_5 + D_6 + D_7$

$Y_0 \Rightarrow \bar{Y}_2 \cdot A + Y_2 C$

↳ $\left[D_1(\overline{D_2 + D_3}) + D_3(D_2 + D_3)\right] \cdot \left[\overline{D_4 + D_5 + D_6 + D_7}\right]$

$+ \left[D_5(\overline{D_6 + D_7}) + D_7(D_6 + D_7)\right]\left[D_4 + D_5 + D_6 + D_7\right]$

expanding ⇒ $(D_1 \cdot \bar{D}_2 \cdot \bar{D}_3 + D_3 \cdot D_2 + D_3)(\bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7) + (D_5 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_6 \cdot D_7 + D_7) \cdot (D_4 + D_5 + D_6 + D_7)$

$D_1 \cdot \bar{D}_2 \cdot \bar{D}_3 + D_3$

↳ $(D_1 \bar{D}_2 + D_3)(\bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7) + (D_5 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_7)(D_4 + D_5 + D_6 + D_7)$

$[\bar{D}_1 \cdot \bar{D}_2 \cdot \bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7] + [D_4 D_5 \cdot \bar{D}_6 \cdot \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_7]$

$\Rightarrow \bar{D}_6 \bar{D}_7 [D_1 \bar{D}_2 \bar{D}_4 \bar{D}_5 + D_3 \cdot \bar{D}_4 \bar{D}_5 + D_4 \cdot D_5 + D_5] + D_7$

$\Rightarrow \bar{D}_6 \bar{D}_7 [\bar{D}_5 (D_1 \cdot \bar{D}_2 \cdot \bar{D}_4 + D_3 \cdot \bar{D}_4) + D_5] + D_7$

$\Rightarrow \bar{D}_6 \cdot \bar{D}_7 [D_1 \bar{D}_2 \cdot \bar{D}_4 + D_5 D_4 + D_5] + D_7$

$\Rightarrow D_7 [D_1 \bar{D}_2 \cdot \bar{D}_4 \bar{D}_6 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_6 + D_5 \cdot \bar{D}_6] + D_7$

$Y_0 \Leftarrow D_1 \cdot \bar{D}_2 \cdot \bar{D}_4 \cdot \bar{D}_6 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_6 + D_5 \cdot \bar{D}_6 + D_7$

$Y_1 = \overline{Y_2 \cdot B} + Y_2 \cdot D$

$\hookrightarrow \overline{(D_4 + D_5 + D_6 + D_7) \cdot (D_2 + D_3)} + D_4 + D_5 + D_6 + D_7 (D_6 + D_7)$

$Y_1 = (\bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7)(D_2 + D_3) + (D_4 D_6 + D_5 \cdot D_6 + D_6 + D_6 \cdot D_7 + D_4 D_7 + D_5 \cdot D_7 + D_6 D_7 + D_7)$

$Y_1 = D_2 \cdot \bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_5 \cdot \bar{D}_6 \cdot \bar{D}_7 + D_6 (D_4 + D_5 + 1 + D_7) + D_7 (D_4 + D_5 + D_6 + 1)$

$Y_1 = \bar{D}_6 \cdot \bar{D}_7 (D_2 \cdot \bar{D}_4 \cdot \bar{D}_5 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_5) + D_6 + D_7$

$Y_1 \Leftarrow Y_1 = \overline{D_6 + D_7} (D_2 \cdot \bar{D}_4 \cdot \bar{D}_5 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_5) + (D_6 + D_7)$

$Y_1 = D_2 \cdot \bar{D}_4 \cdot \bar{D}_5 + D_3 \cdot \bar{D}_4 \cdot \bar{D}_5 + D_6 + D_7$

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Y_2	Y_1	Y_0	V
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	1	x	0	0	1	1
0	0	0	0	0	1	x	x	0	1	0	1
0	0	0	0	1	x	x	x	0	1	1	1
0	0	0	1	x	x	x	x	1	0	0	1
0	0	1	x	x	x	x	x	1	0	1	1
0	1	x	x	x	x	x	x	1	1	0	1
1	x	x	x	x	x	x	x	1	1	1	1
x	x	x	x	x	x	x	x	0	0	0	0

Exercise 3.5 ⇒ Implementation of the Function F(A, B, C)

From the table;

$F(A,B,C) \Rightarrow \sum m(0,1,5,6,7)$

$F(A,B,C) = \bar{B}$ @ $A=0$

and also $B+C$ @ $A=1$

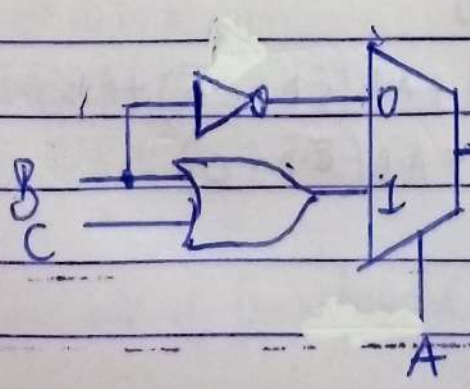
A	B	C	F	
0	0	0	1	$\bar{B}$
0	0	1	1	
0	1	0	0	
0	1	1	0	
1	0	0	0	$B+C$
1	0	1	1	
1	1	0	1	
1	1	1	1	

To get the implementation

$F(A,B,C) \Rightarrow (0,B,C)$ or $(1,B,C)$

$\Rightarrow (0,B,C) + (1,B,C)$ @ $A=0$ $\bar{B}$; @ $A=1$ $B+C$

$\bar{B} + B+C$



2:1 Multiplexer.

Exercise 3.6: 4:1 Multiplexer

Y/F

Logic Gate (A) ⇒ $\bar{S}_1 \cdot \bar{S}_0 \cdot D_0 \cdot E$

Logic Gate (B) ⇒ $\bar{S}_1 \cdot S_0 \cdot D_1 \cdot E$

Logic Gate (C) ⇒ $S_1 \cdot \bar{S}_0 \cdot D_2 \cdot E$

Logic Gate (D) ⇒ $S_1 \cdot S_0 \cdot D_3 \cdot E$

$Y = A + B + C + D$

$Y = \bar{S}_1 \cdot \bar{S}_0 \cdot D_0 E + \bar{S}_1 \cdot S_0 \cdot D_1 \cdot E + S_1 \cdot \bar{S}_0 \cdot D_2 \cdot E + S_1 \cdot S_0 \cdot D_3 E$

E	S_1	S_0	Y
0	X	X	0
1	0	0	D_0
1	0	1	D_1
1	1	0	D_2
1	1	1	D_3

→ When enabler is 0, all other inputs are give to be inactive; hence, do-not care situation, leading to a Low output

→ When enabler is 1, output then depends on S_1 & S_0

Use a 4-to-1 multiplexer and logic gates to implement the function:

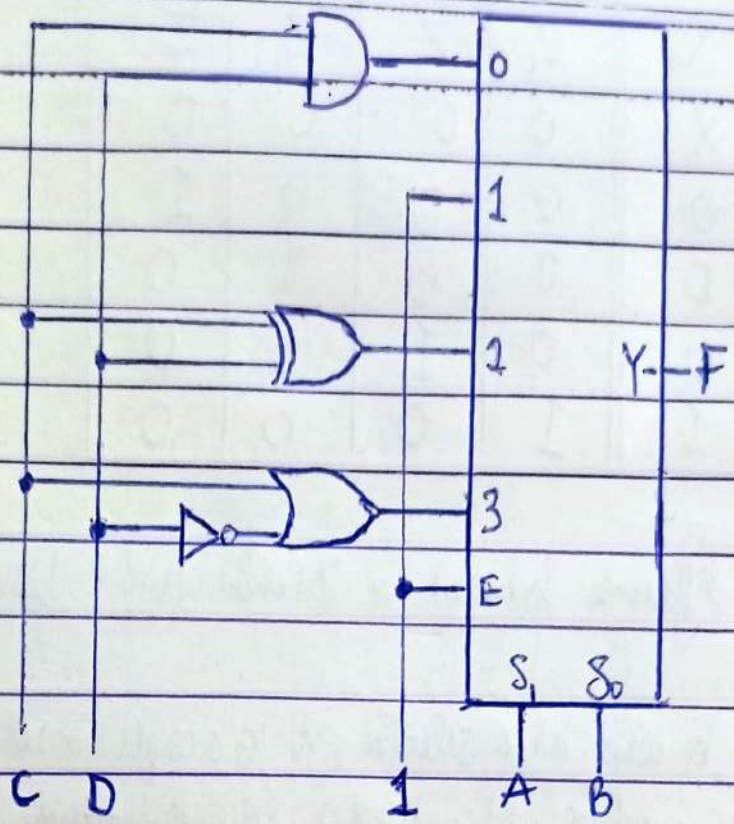
$F(A,B,C,D) = \Sigma m(3,4,5,6,7,9,10,12,14,15)$

⇒ $\bar{A} \cdot B \cdot \bar{C} \cdot D + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D} + \bar{A} \cdot B \cdot \bar{C} \cdot D + \bar{A} \cdot B \cdot C \cdot \bar{D} + \bar{A} \cdot B \cdot C \cdot D + A \cdot \bar{B} \cdot \bar{C} \cdot D + A \cdot \bar{B} \cdot C \cdot \bar{D}$
$+ A \cdot B \cdot \bar{C} \cdot \bar{D} + A \cdot B \cdot C \cdot \bar{D} + A \cdot B \cdot C \cdot D$

⇒ $\bar{A} \cdot \bar{B} \cdot C \cdot D + \bar{A} \cdot B (\bar{C} \cdot \bar{D} + \bar{C} \cdot D + C \cdot \bar{D} + C \cdot D) + A \cdot \bar{B} (\bar{C} \cdot D + C \cdot \bar{D}) + A \cdot B (\bar{C} \cdot \bar{D} + C \cdot \bar{D} + C \cdot D)$

$\bar{A} \cdot \bar{B} \cdot C \cdot D + \bar{A} \cdot B (1 + 1) + A \cdot \bar{B} (C \oplus D) + A \cdot B (\bar{C} \cdot \bar{D} + C)$

⇒ $\bar{A} \cdot \bar{B} \cdot C \cdot D + \bar{A} \cdot B + A \cdot \bar{B}$



Karnaugh Map:

A \ B	0	1
0	C.D	1
1	C⊕D	C+D

Exercise 8.7 :- Demultiplexer / Decoder.

(a) Draw up the truth table for the decoder in Fig 3.65(a)

$Z_0 \Rightarrow \overline{S_1} \cdot \overline{S_0} \cdot E$

$Z_1 \Rightarrow \overline{S_1} S_0 \cdot E$

$Z_2 \Rightarrow S_1 \overline{S_0} \cdot E$

$Z_3 \Rightarrow S_1 S_0 \cdot E$

A two-out-of four decoder 2:4 DEC

E	S_1	S_0	Z_3	Z_2	Z_1	Z_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

(b) What is the difference between a Decoder And Demultiplexer

A Decoder can be used as a selector, it accepts a set of inputs that represents a binary number and activates only the output that corresponds to that input number.

Formular; n-out-of-2^n decoder ; n input data and 2^n outputs

A Demultiplexer ;also called Data distributors (DEMUX)
It takes a single input and distributes it over several outputs
Formular 1-to-2^n demultiplexer : 1 data input, n Select inputs and 2^n outputs-

(c) Analyze the logic circuit shown in Fig 3.65(g), determine the logic equations for the output $Z(A,B,C,D)$ with $i = 0,1,2,3$.

$$Z_0 = \overline{S_1} \cdot \overline{S_0} \cdot E$$

$$Z_1 = \overline{S_1} \cdot S_0 \cdot E$$

$$Z_2 = S_1 \cdot \overline{S_0} \cdot E$$

$Z_i = S_1 \cdot S_0 \cdot E$

Now to Analyze 4:1 MUX connected to a 2:4 DEC keeping in mind we obtain our logic function equation for; $Z_i (A, B, C, D)$ $i = 0, 1, 2, 3$

4:1 MUX:

Output $Y = \sum [0 + 1 + 2 + 3] E$

$0 \Rightarrow \bar{S}_1 \bar{S}_0 \cdot (C + D)$

$1 \Rightarrow \bar{S}_1 \cdot S_0 (2)$

$2 \Rightarrow S_1 \bar{S}_0 \cdot C$

$3 \Rightarrow S_1 \cdot S_0 \cdot \bar{C} \cdot \bar{D}$

$Y = [\bar{S}_1 \cdot \bar{S}_0 (C + D) + \bar{S}_1 \cdot \bar{S}_0 \cdot C + S_1 S_0 \cdot C \cdot \bar{D}] E$

@ $E = 1$ $S_0 = A$ $S_1 = B$

$Y = [\bar{B} \cdot A (C + D) + B \cdot A \cdot C + B \bar{A} \cdot \bar{C} \cdot D]$

$Y = [A \cdot \bar{B} C + A \bar{B} \cdot D + A \cdot B \cdot C + \bar{A} \cdot B \bar{C} \cdot \bar{D}]$

$Y = [A \cdot C + A \cdot \bar{B} \cdot D + \bar{A} B \bar{C} \cdot \bar{D}]$

Final 2:4 DEC;

To get E; $Y = E$ @

$E = [AC + A \bar{B} \cdot D + \bar{A} B \cdot \bar{C} \cdot \bar{D}]$, $S_1 = \bar{A}$, $S_0 = \bar{A} \oplus B$

$Z_0 = \bar{S}_1 \bar{S}_0 \cdot E$

$\hookrightarrow A \cdot \bar{A} \oplus B \cdot (AC + A \bar{B} \cdot D + \bar{A} B \bar{C} \cdot \bar{D})$

$\hookrightarrow A (\bar{A} \cdot B + A \bar{B}) (AC + A \cdot \bar{B} \cdot D + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D})$

$\hookrightarrow A \cdot B (A \cdot C + A \cdot \bar{B} \cdot D + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D})$

$Z_0 \hookrightarrow A \cdot \bar{B} \cdot C + A \cdot \bar{B} \cdot D$

$Z_1 = \bar{S}_1 S_0 E$

$= A \cdot (\bar{A} \oplus B) \cdot E$

$\Rightarrow A \cdot [A \cdot B + \bar{A} \cdot \bar{B}] \cdot E$

$\Rightarrow A \cdot B (A \cdot C + A \cdot \bar{B} \cdot D + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D})$

$Z_1 \Rightarrow A \cdot B \cdot C$

$Z_2 = S_1 \cdot \bar{S}_0 \cdot E$

$\Rightarrow \bar{A} \cdot \overline{\bar{A} \oplus B} \cdot E$

$\Rightarrow \bar{A} \cdot (\bar{A} \cdot B + A \bar{B}) \cdot E$

$\Rightarrow \bar{A} \cdot B \cdot E$

$\Rightarrow \bar{A} B (A \cdot C + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D} + A \cdot \bar{B} \cdot D)$

$Z_2 \Rightarrow \bar{A} B \bar{C} \bar{D}$

$Z_3 \Rightarrow S_1 S_0 \bar{E}$

$\Rightarrow \bar{A} \cdot \overline{\bar{A} \oplus B} \cdot \bar{E}$

$\Rightarrow \bar{A} (A \cdot B + \bar{A} \bar{B}) \cdot \bar{E}$

$\Rightarrow 0 + \bar{A} \bar{B} \cdot \bar{E}$

$\Rightarrow \bar{A} \bar{B} (A \cdot C + \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D} + A \cdot \bar{B} \cdot D)$

$Z_3 \Rightarrow 0$

Ans

$Z_0 = A \bar{B} C + A \cdot \bar{B} D$

$Z_1 = A \cdot B \cdot C$

$Z_2 = \bar{A} \cdot B \cdot \bar{C} \cdot \bar{D}$

$Z_3 = 0$

Exercise 3.8 :- Implementation of the function $F(A,B,C,D)$

Implementing function $F(A,B,C,D) = A \oplus B \oplus C \oplus D$ using 4:1 multiplexer and logic gates to be determined.

[93]

Expansion with Shannon's theorem

$$F(x_0, x_1, \ldots, x_i, \ldots, x_{n-1}) = \bar{x}_i \cdot F(x_0, x_1, \ldots, 0, \ldots, x_{n-1}) + x_i \cdot F(x_0, x_1, \ldots, 1, \ldots, x_{n-1})$$

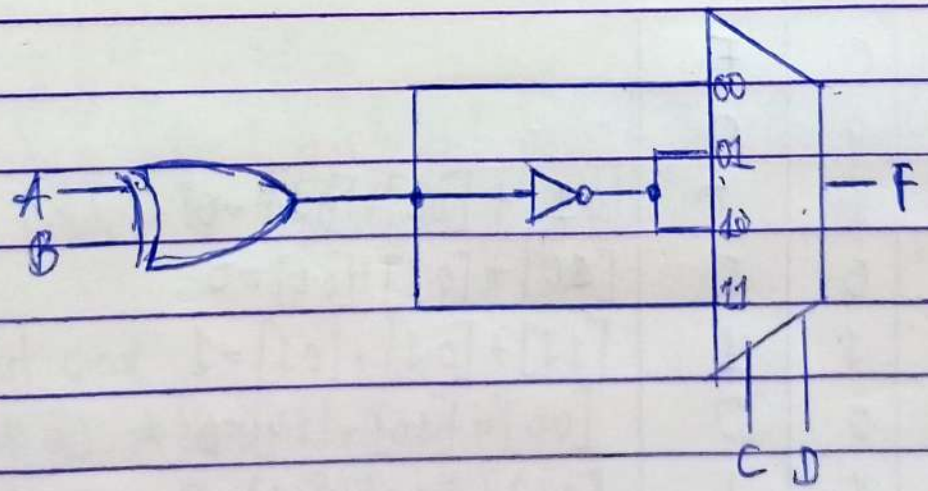
$F(A,B,C,D) = A \oplus B \oplus C \oplus D$

$(A \oplus B \oplus 0 \oplus 0)\bar{C}\cdot\bar{D} + (A \oplus B \oplus 0 \oplus 1)\bar{C}\cdot D + (A \oplus B \oplus 1 \oplus 0)C\cdot\bar{D} + (A \oplus B \oplus 1 \oplus 1)C\cdot D$

$[A \oplus B \oplus (\bar{0}\cdot 0 + 0\cdot\bar{0})]\bar{C}\cdot\bar{D} + [A \oplus B \oplus (\bar{0}\cdot 1 + 0\cdot\bar{1})]\bar{C}\cdot D + [A \oplus B \oplus (\bar{1}\cdot 0 + 1\cdot\bar{0})]C\cdot\bar{D} + [A \oplus B \oplus (\bar{1}\cdot 1 + 1\cdot\bar{1})]C\cdot D$

$[(A \oplus B)\cdot\bar{0} + (\overline{A \oplus B})\cdot 0]\bar{C}\cdot\bar{D} + [(A \oplus B)\cdot\bar{1} + (\overline{A \oplus B})\cdot 1]\bar{C}\cdot D + [(A \oplus B)\cdot\bar{1} + (\overline{A \oplus B})\cdot 1]C\cdot\bar{D} + [(A \oplus B)\cdot\bar{0} + (\overline{A \oplus B})\cdot 0]C\cdot D$

$[A \oplus B]\bar{C}\cdot\bar{D} + [A \oplus B]\bar{C}\cdot D + [\overline{A \oplus B}]C\cdot\bar{D} + [A \oplus B]C\cdot D$



Implementation using a 4:1 multiplexer =

Exercise 3.9 - Majority Function Of Three Variables

Majority function, majority of functions of three variables means two or three of the functions take a logic level (High = 1)

$$F(A,B,C) = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$= BC(\bar{A}+A) + A\bar{B}C + AB\bar{C}$$

$$= BC + A\bar{B}C + AB\bar{C}$$

$$\Rightarrow B(C + A\bar{C}) + A\bar{B}C$$

$$B(C+A) + A\bar{B}C$$

$$BC + AB + A\bar{B}C$$

$$BC + A(B + \bar{B}C)$$

$$F = BC + AB + AC$$

(A) Truth Table of the function F

Using a Bit system

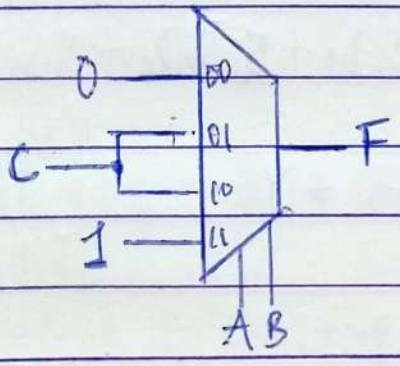
A	B	C	F	
0	0	0	0	
0	0	1	0	$[0\cdot1] + [0\cdot0] + [0\cdot1] = 0$
0	1	0	0	$[1\cdot0] + [0\cdot1] + [0\cdot0] = 0$
0	1	1	1	$[1\cdot1] + [0\cdot1] + [0\cdot1] = 1$
1	0	0	0	$[0\cdot0] + [1\cdot0] + [1\cdot0] = 0$
1	0	1	1	$[0\cdot1] + [1\cdot0] + [1\cdot1] = 1$
1	1	0	1	$[1\cdot0] + [1\cdot1] + [1\cdot0] = 1$
1	1	1	1	$[1\cdot1] + [1\cdot1] + [1\cdot1] = 1$

Meaning F will be active if any two or more input is High

(b) Determine and simplify the boolean equation for the function F,

$F = B \cdot C + A \cdot B + A \cdot C$

(c) Implement the function F using a 4:1 multiplexer



Exercise 3-10 ⇒ 8-to-1 Multiplexer

Verifying Fig 3-66 works as a 8-to-1 multiplexer.

A typical 8-to-1 multiplexer

$$F = \bar{S}_2 \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_2 \bar{S}_1 S_0 D_1 + \bar{S}_2 S_1 \bar{S}_0 D_2 + \bar{S}_2 S_1 S_0 D_3 + S_2 \bar{S}_1 \bar{S}_0 D_4 + S_2 \bar{S}_1 S_0 D_5 + S_2 S_1 \bar{S}_0 D_6 + S_2 S_1 S_0 D_7 \quad \text{— eqn [3-10]}$$

for the first circuit

First block (a) ⇒ $\bar{S}_0 \cdot D_0 + S_0 D_1 \Rightarrow A$

(b) ⇒ $\bar{S}_0 D_2 + S_0 D_3 \Rightarrow B$

(c) ⇒ $\bar{S}_0 \cdot D_4 + S_0 \cdot D_5 \Rightarrow C$

(d) ⇒ $\bar{S}_0 D_6 + S_0 \cdot D_7 \Rightarrow D$

Main block ⇒ $\bar{S}_2 \cdot \bar{S}_1 A + \bar{S}_2 \cdot S_1 \cdot B + S_2 \bar{S}_1 \cdot C + S_2 \cdot S_1 \cdot D$

↳ $\bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_0 + \bar{S}_2 \cdot \bar{S}_1 S_0 D_1 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_2 + \bar{S}_2 S_1 \cdot S_0 \cdot D_3 + S_2 \bar{S}_1 \bar{S}_0 \cdot D_4$

$\cdots D \cdots \bar{S}_2 \bar{S}_1 \bar{S}_0 D_5 + \bar{S}_2 S_1 \bar{S}_0 D_6 + S_2 S_1 S_0 D_7$

This is equal to the eqn [3.10], hence it is a 8:1 MUX

Second Circuit;

First block (A); $\bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3 = A$

Second block (B); $\bar{S}_1 \bar{S}_0 D_4 + \bar{S}_1 S_0 D_5 + S_1 \bar{S}_0 D_6 + S_1 S_0 D_7 = B$

Last block $A\bar{S}_2 + BS_2 = Y$

$\therefore$ ~~$Y = \bar{S}_2 \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_2 S_1$~~

$Y = \bar{S}_2 (\bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3) + S_2 (S_1 \bar{S}_0 D_4 + S_1 S_0 D_5 + \bar{S}_1 \bar{S}_0 D_6 + S_1 S_0 D_7)$

$Y = \bar{S}_2 \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_2 \bar{S}_1 S_0 D_1 + \bar{S}_2 S_1 \bar{S}_0 D_2 + \bar{S}_2 S_1 S_0 D_3 + S_2 \bar{S}_1 \bar{S}_0 D_4 + S_2 \bar{S}_1 S_0 D_5 + S_2 S_1 \bar{S}_0 D_6 + S_2 S_1 S_0 D_7$

which is also equal to the eqn [3.12]

hence it does act as a 8:1 MUX

S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

Use an 8-to-1 MUX and logic gates to implement

a) $F(A,B,C,D,E) = \sum m(0,1,2,4,5,6,7,13,14,20,21,22,28,29,30,31)$

↳ $\bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}E + \bar{A}B\bar{C}\bar{D}\bar{E} + \bar{A}B\bar{C}\bar{D}E + \bar{A}B\bar{C}D\bar{E} + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}CD\bar{E} + A\bar{B}C\bar{D}\bar{E} + A\bar{B}CD\bar{E} + AB\bar{C}\bar{D}\bar{E} + A\bar{B}CD\bar{E} + AB\bar{C}D\bar{E} + AB\bar{C}D\bar{E} + A\bar{B}CD\bar{E} + AB\bar{C}D\bar{E}$

5 variables

$2^5 - 1 = 31$ possible inputs

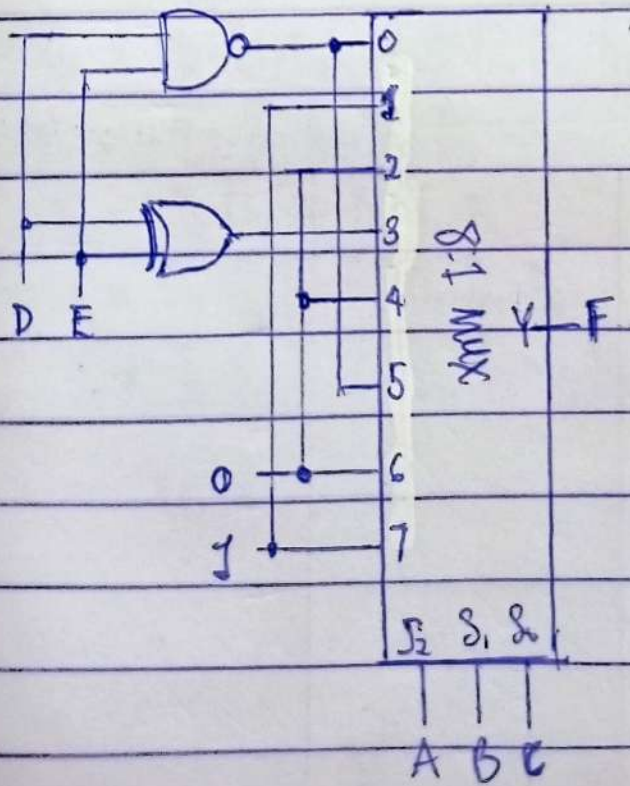
↳ $(\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C})(\bar{D}\bar{E} + \bar{D}E + D\bar{E}) + (\bar{A}B\bar{C} + ABC)(\bar{D}\bar{E} + \bar{D}E + D\bar{E} + DE) + \bar{A}B\bar{C}(\bar{D}E + D\bar{E})$

First ⇒ $\bar{D}\bar{E} + \bar{D}E + D\bar{E} \Rightarrow \bar{E}(\bar{D}+D) + \bar{D}E \Rightarrow \bar{E} + \bar{D}E \Rightarrow \bar{D} + \bar{E}$

$\bar{D}\bar{E} + \bar{D}E + D\bar{E} + DE \Rightarrow \bar{D}(\bar{E}+E) + D(\bar{E}+E) \Rightarrow \bar{D}+D = 1$

$\bar{D}E + D\bar{E} \Rightarrow D \oplus E$

↳ $(\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C})(\bar{D}+\bar{E}) + (A\bar{B}C + ABC)(1) + \bar{A}BC(D \oplus E)$



AB \ C	
	1
0	D⊕E
0	$\bar{D}+\bar{E}$
0	1

Exercise 3.11 – Implementation Of A Logic Function Based On A Decoder

Implement the logic function:

$F(A,B,C,D) = \sum m(1,3,7,9,15)$

using a 3-to-8 decoder with enable input $\overline{EN}$, a 5 input NAND gate and 2-input NAND gates. The decoder has active-low outputs.

$F = \bar{A}\cdot\bar{B}\cdot\bar{C}\cdot D + \bar{A}\cdot\bar{B}\cdot C\cdot D + \bar{A}\cdot B\cdot C\cdot D + A\bar{B}\cdot\bar{C}\cdot D + A\cdot B\cdot C\cdot D$

3-to-8 Decoder;

$Y_0 = \bar{S}_2\cdot\bar{S}_1\cdot\bar{S}_0\cdot\overline{EN}$ $Y_1 = \bar{S}_2\cdot\bar{S}_1 S_0\cdot\overline{EN}$ $Y_2 = \bar{S}_2 S_1\cdot\bar{S}_0\cdot\overline{EN}$ $Y_3 = \bar{S}_2 S_1 S_0\cdot\overline{EN}$

$Y_4 = S_2\bar{S}_1\bar{S}_0\cdot\overline{EN}$ $Y_5 = S_2\bar{S}_1 S_0\cdot\overline{EN}$ $Y_6 = S_2 S_1\bar{S}_0\cdot\overline{EN}$ $Y_7 = S_2 S_1 S_0\cdot\overline{EN}$

Comparing and Substituting

$Y_0 = \bar{S}_2\cdot\bar{S}_1\cdot\bar{S}_0\cdot\overline{EN} = \bar{A}\cdot\bar{B}\cdot\bar{C}\cdot D$

$Y_1 = \bar{S}_2\cdot\bar{S}_1\cdot S_0\cdot\overline{EN} = \bar{A}\cdot\bar{B}\cdot C\cdot D$

$Y_3 = \bar{S}_2 S_1 S_0\cdot\overline{EN} = \bar{A}\cdot B\cdot C\cdot D$

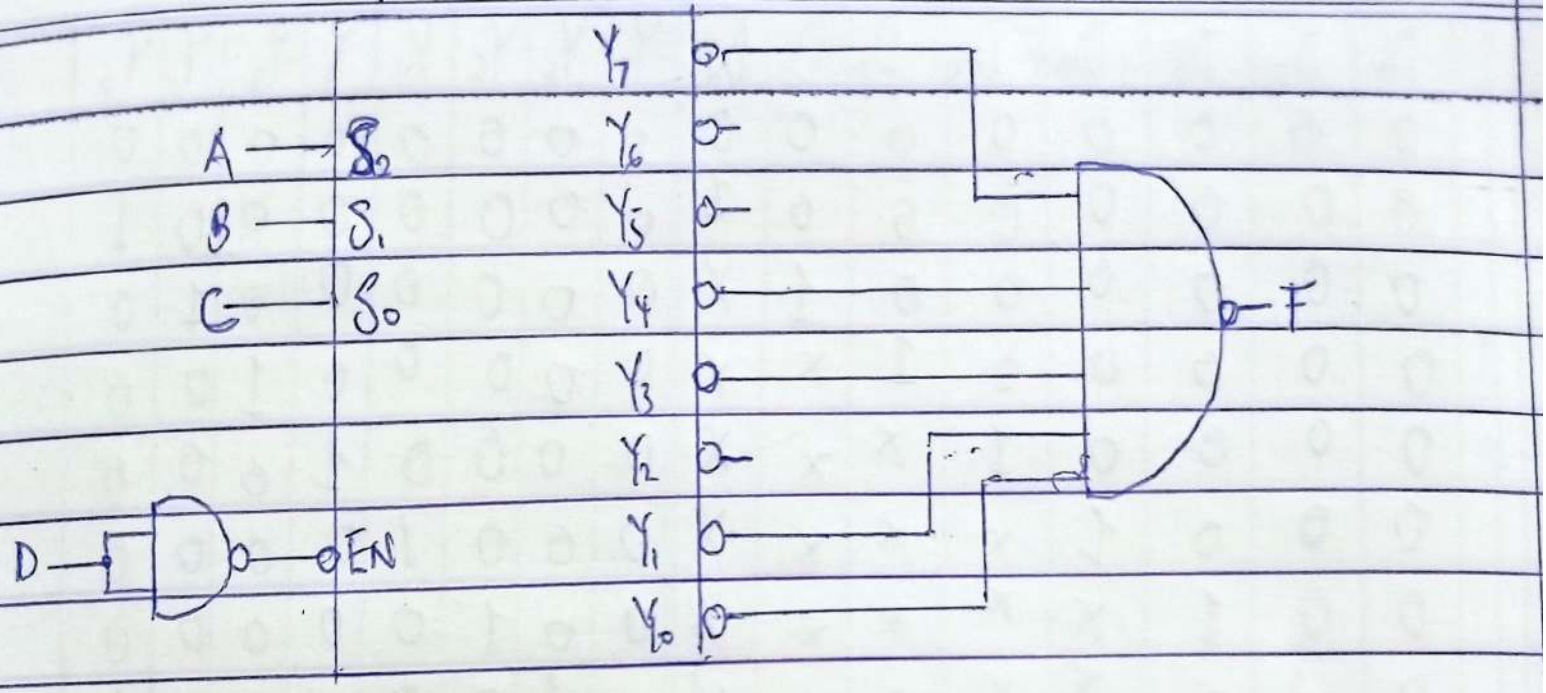
$Y_4 = S_2\bar{S}_1\bar{S}_0\cdot\overline{EN} = A\bar{B}\bar{C}D$

$Y_7 = S_2 S_1 S_0\cdot\overline{EN} = ABCD$

$\therefore F = Y_0 + Y_1 + Y_3 + Y_4 + Y_7$

For A NAND

$F = \overline{Y_0}\cdot\overline{Y_1}\cdot\overline{Y_3}\cdot\overline{Y_4}\cdot\overline{Y_7}$



Exercise 3.12 : Analysis of A Logic Circuit

Determine the logic equation for each of the outputs $Y_i (i=0,1,2,3,4,5,6,7)$ of the logic circuit shown in Figure 3-67

$Y_7 = D_7$

$Y_6 = \bar{D}_7 \cdot D_6$

$Y_5 = \bar{D}_7 \cdot \bar{D}_6 \cdot D_5$

$Y_4 = \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_5 \cdot D_4$

$Y_3 = \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_5 \cdot \bar{D}_4 \cdot D_3$

$Y_2 = \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_5 \cdot \bar{D}_4 \cdot \bar{D}_3 \cdot D_2$

$Y_1 = \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_5 \cdot \bar{D}_4 \cdot \bar{D}_3 \cdot \bar{D}_2 \cdot D_1$

$Y_0 = \bar{D}_7 \cdot \bar{D}_6 \cdot \bar{D}_5 \cdot \bar{D}_4 \cdot \bar{D}_3 \cdot \bar{D}_2 \cdot \bar{D}_1 \cdot D_0$

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	X	0	0	0	0	0	0	1	0
0	0	0	0	0	1	X	X	0	0	0	0	0	1	0	0
0	0	0	0	1	X	X	X	0	0	0	0	1	0	0	0
0	0	0	1	X	X	X	X	0	0	0	1	0	0	0	0
0	0	1	X	X	X	X	X	0	0	1	0	0	0	0	0
0	1	X	X	X	X	X	X	0	1	0	0	0	0	0	0
1	X	X	X	X	X	X	X	1	0	0	0	0	0	0	0

Exercise 3.13; Design of a Multiplier for 2 Bit Binary Numbers

Implementation of a multiplier;

- input : $X = X_1X_0$ & $Y = Y_1Y_0$
- output : $Z = X \cdot Y$ & $Z = Z_3Z_2Z_1Z_0$

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	X	0	0	0	0	0	0	1	0
0	0	0	0	0	1	X	X	0	0	0	0	0	1	0	0
0	0	0	0	1	X	X	X	0	0	0	0	1	0	0	0
0	0	0	1	X	X	X	X	0	0	0	1	0	0	0	0
0	0	1	X	X	X	X	X	0	0	1	0	0	0	0	0
0	1	X	X	X	X	X	X	0	1	0	0	0	0	0	0
1	X	X	X	X	X	X	X	1	0	0	0	0	0	0	0

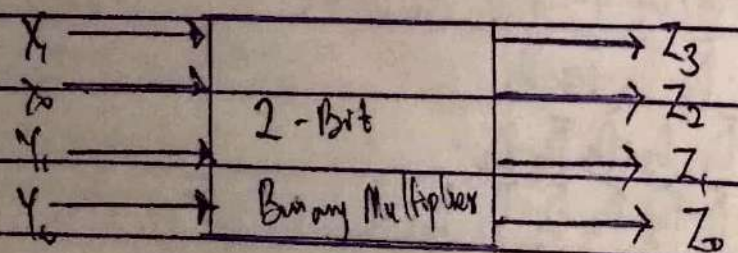
Truth table for Ex 3.12

Example 3.13; Design Of A Multiplier For 2 Bit Binary Words

Implementation of a multiplier;

input - $X = X_1X_0$ & $Y = Y_1Y_0$

output ; $Z = X \cdot Y$ - & $Z = Z_3Z_2Z_1Z_0$



A/ Truth Table for the Multiplier ; a 2-bit words multiplier

X·Y=Z	X		Y		Z			
	X_1	X_0	Y_1	Y_0	Z_3	Z_2	Z_1	Z_0
0·0=0	0	0	0	0	0	0	0	0
0·1=0	0	0	0	1	0	0	0	0
0·2=0	0	0	1	0	0	0	0	0
0·3=0	0	0	1	1	0	0	0	0
1·0=0	0	1	0	0	0	0	0	0
1·1=1	0	1	0	1	0	0	0	1
1·2=2	0	1	1	0	0	0	1	0
1·3=3	0	1	1	1	0	0	1	1
2·0=0	1	0	0	0	0	0	0	0
2·1=2	1	0	0	1	0	0	1	0
2·2=4	1	0	1	0	0	1	0	0
2·3=6	1	0	1	1	0	1	1	0
3·0=0	1	1	0	0	0	0	0	0
3·1=3	1	1	0	1	0	0	1	1
3·2=6	1	1	1	0	0	1	1	0
3·3=9	1	1	1	1	1	0	0	1

B/ Boolean Expressions for each of the outputs $Z_i (i = 0, 1, 2, 3)$

from the table we see

$Z_3 \Rightarrow X_1 \cdot X_0 \cdot Y_1 \cdot Y_0$ (no need for a K-map)

$Z_2 \Rightarrow X_1 \cdot \bar{X}_0 \cdot Y_1 \cdot Y_0 + X_1 \cdot \bar{X}_0 \cdot Y_1 \cdot \bar{Y}_0 + X_1 \cdot X_0 \cdot Y_1 \cdot Y_0$

$Z_1 \Rightarrow X_1 \cdot X_0 \cdot Y_1 \cdot \bar{Y}_0 + \bar{X}_1 \cdot X_0 \cdot Y_1 \cdot Y_0 + X_1 \cdot X_0 \cdot \bar{Y}_1 \cdot \bar{Y}_0 + X_1 \cdot \bar{X}_0 \cdot Y_1 \cdot Y_0 + X_1 \cdot X_0 \cdot \bar{Y}_1 \cdot Y_0 + X_1 \cdot X_0 \cdot Y_1 \cdot \bar{Y}_0$

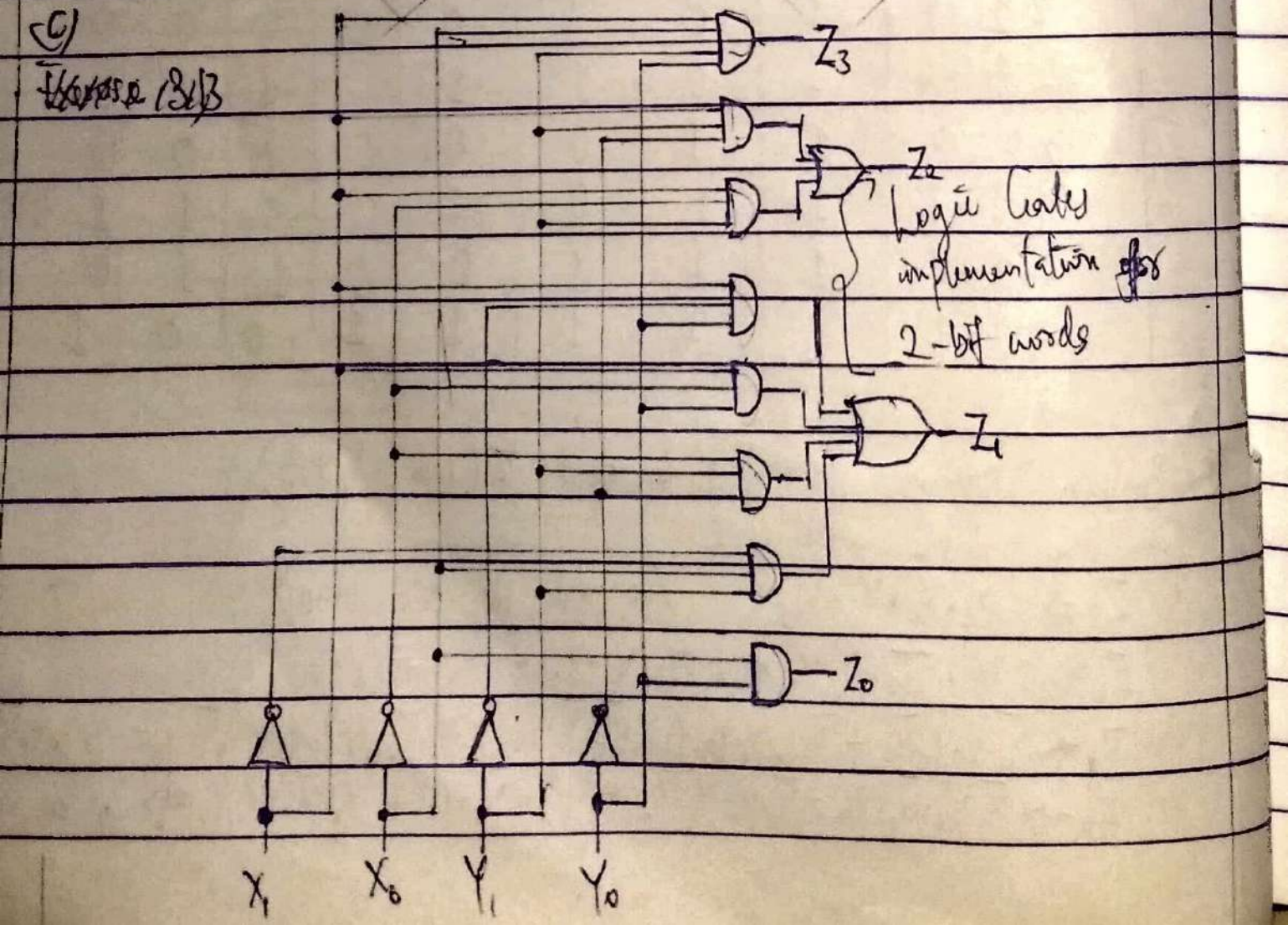
$Z_0 \Rightarrow X_1 \cdot X_0 \cdot Y_1 \cdot Y_0 + \bar{X}_1 \cdot X_0 \cdot Y_1 \cdot Y_0 + X_1 \cdot X_0 \cdot \bar{Y}_1 \cdot Y_0 + X_1 \cdot X_0 \cdot Y_1 \cdot Y_0 \Rightarrow X_0 \cdot Y_0$

Truth Table for This Circuit

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	x	0	0	0	0	0	0	1	0
0	0	0	0	0	1	x	x	0	0	0	0	0	1	0	0
0	0	0	0	1	x	x	x	0	0	0	0	1	0	0	0
0	0	0	1	x	x	x	x	0	0	0	1	0	0	0	0
0	0	1	x	x	x	x	x	0	0	1	0	0	0	0	0
0	1	x	x	x	x	x	x	0	1	0	0	0	0	0	0
1	x	x	x	x	x	x	x	1	0	0	0	0	0	0	0

c)

Inverse (3/3)

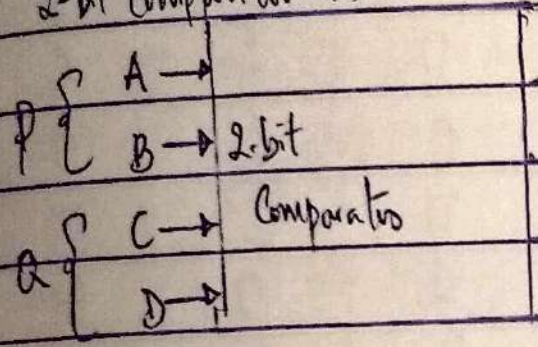


Exercise 3.14: Comparator for 2-Bit Binary Numbers

A comparator compares two binary numbers. P = AB Q = CD

Comparators can be used to determine whether a given number is smaller than, equal to or larger than another number

A 2-bit comparator has



Normal 1-bit comp
$P>Q \Rightarrow P\bar{Q}$
$P=Q = P \odot Q$
$P<Q = \bar{P}Q$

~~$X \Rightarrow P\bar{Q} \Rightarrow A\bar{}$~~

~~Let's write Truth Table for this Comparator:~~

So from the truth table for X we get the equation;

$X = \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D}$

$\Rightarrow \bar{A}B\bar{C}\bar{D} + A\bar{C}(\bar{B}\bar{D} + \bar{B}D + B\bar{D} + BD) + A\bar{B}C\bar{D}$

$\bar{A}B\bar{C}\bar{D} + A\bar{C}(1+1) + A\bar{B}C\bar{D}$

$\bar{A}B\bar{C}\bar{D} + A\bar{C} + A\bar{B}C\bar{D}$

$\bar{A}B\bar{C}\bar{D} + A(\bar{C} + B\bar{C}\bar{D})$

$\bar{A}B\bar{C}\bar{D} + A\bar{C} + A\bar{B}C\bar{D}$

~~$A\bar{C} + \bar{C}(\bar{A}B\bar{D} + A)$~~

~~$A\bar{C}\bar{D} + A\bar{C} + AB\bar{C}\bar{D}$~~

$A\bar{C} + B(\bar{C}\bar{D} + A\bar{C}\bar{D})$

$A\bar{C} + B\bar{C}\bar{D} + AB\bar{D}$

$Y = \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C\bar{D}$

$\Rightarrow \bar{A}\bar{B}\bar{C}D + \bar{A}C(\bar{B}\bar{D} + \bar{B}D + B\bar{D} + BD) + A\bar{B}C\bar{D}$

(a) Truth table for the Comparator;

Inputs				Outputs		
A	B	C	D	X	Y	Z
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	0	1

Cont'd b

$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}C(1+1) + A\bar{B}CD$

$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}C + A\bar{B}CD$

$\bar{A}(\bar{B}\bar{C}\bar{D} + C) + A\bar{B}CD$

$\bar{A}\bar{C} + \bar{A}\bar{B}D + \bar{A}\bar{B}CD$

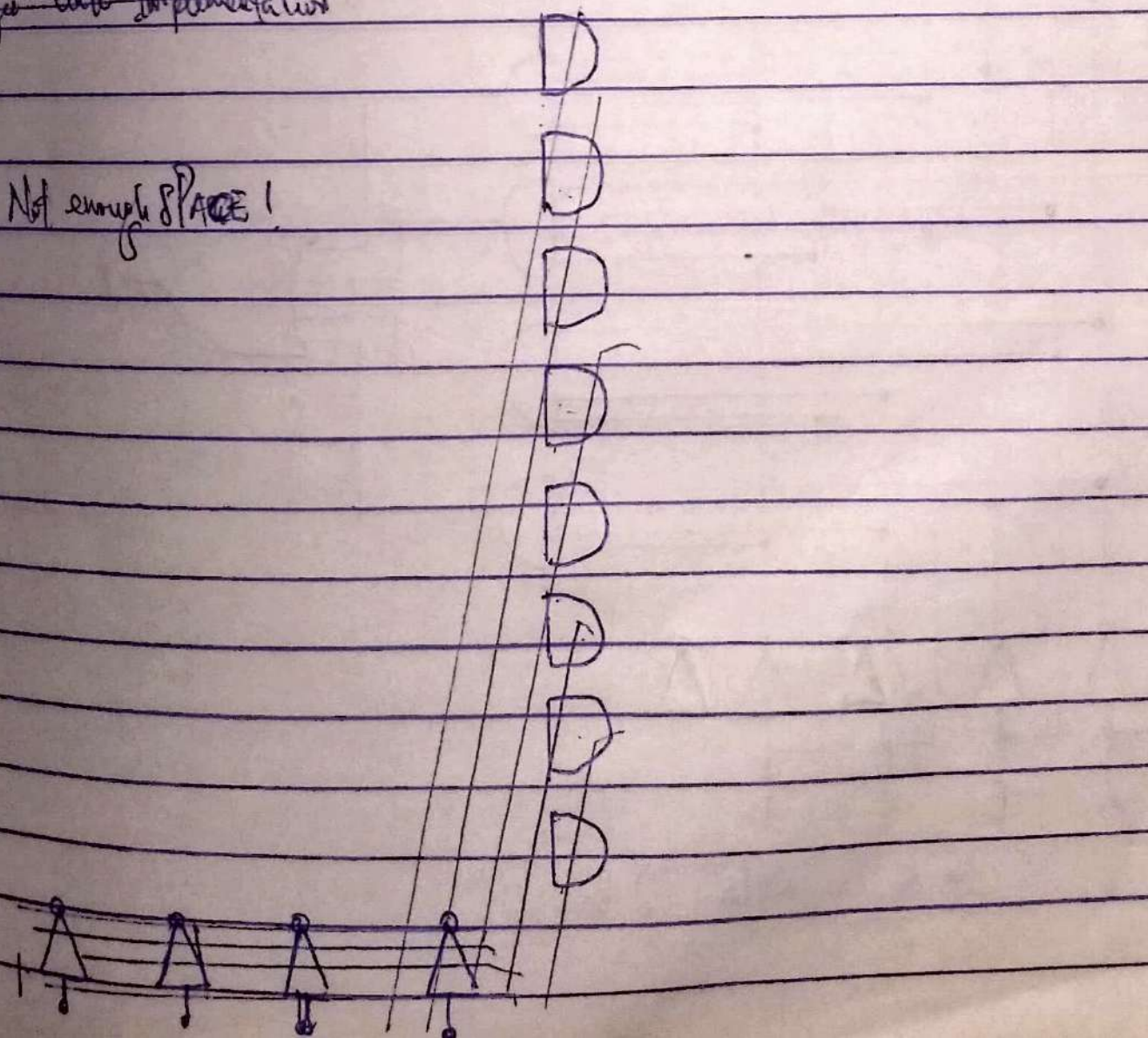
$\bar{A}C + \bar{A}\bar{B}(D + CD)$

$\bar{A}C + \bar{A}\bar{B}D + \bar{B}CD$

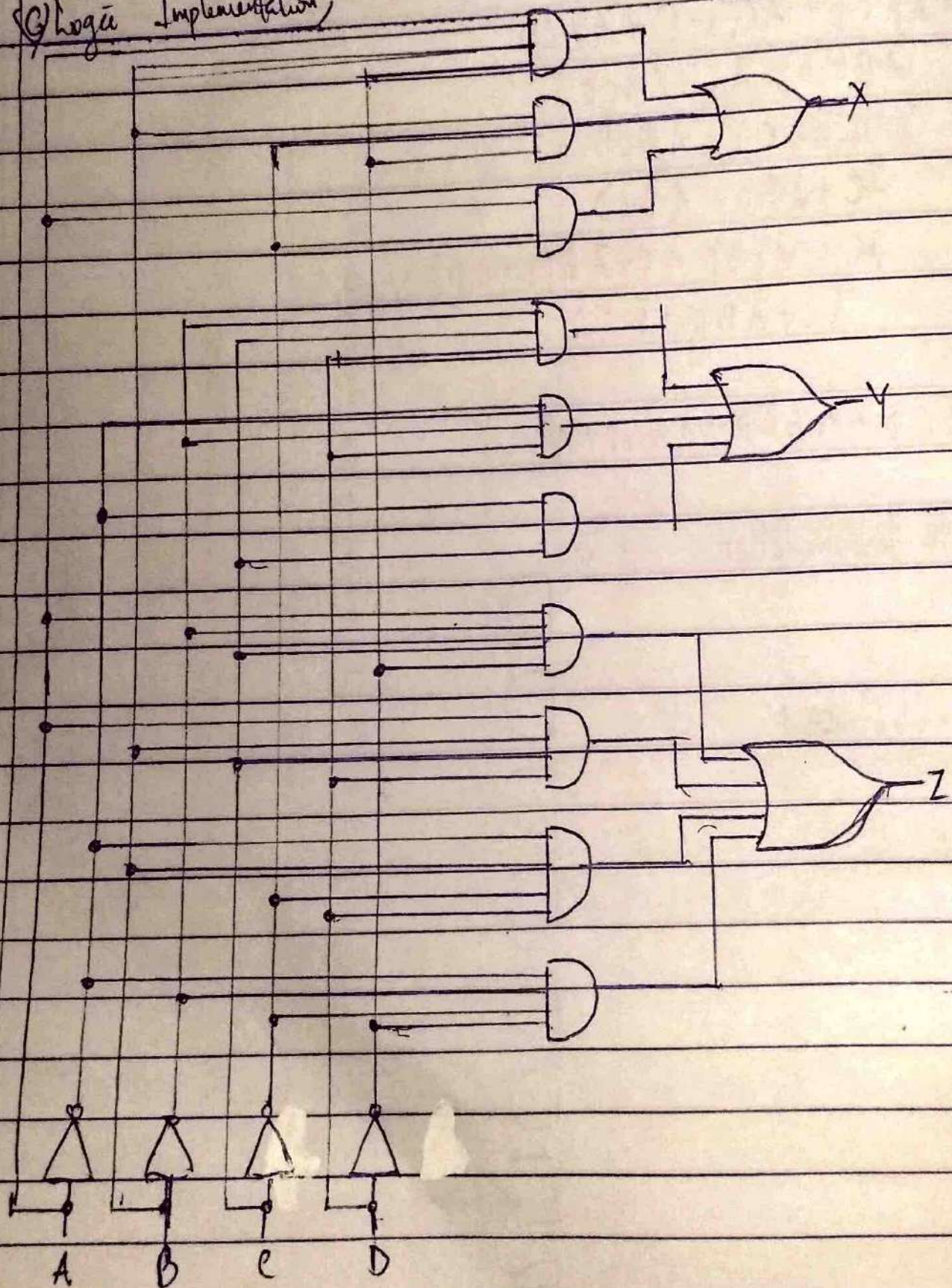
$Z = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + ABCD + A\bar{B}C\bar{D}$

~~Cotlogic Gate Implementation~~

Not enough SPACE!



Logic Implementation



for a Comparator 2-Bit

Exercise 3.5 : BCD-to-7 Segment Decoder

A 4-bit number, A, B, C, D (D is LSB) is applied to the inputs of the decoder supplying the signals a, b, c, d, e, f and g that are used to drive a 7-segment display @ Fig 3.69 generating numbers from 0 to 9.

a) Truth Table For this Decoder (A regular BCD-to-7-Segment Decoder)

Convention makes numbers from 0-9 so the rest are do not care situations

Symbols	BCD-code A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
-	1	0	1	0	x	x	x	x	x	x	x
-	1	0	1	1	x	x	x	x	x	x	x
-	1	1	0	0	x	x	x	x	x	x	x
-	1	1	0	1	x	x	x	x	x	x	x
-	1	1	1	0	x	x	x	x	x	x	x
-	1	1	1	1	x	x	x	x	x	x	x

* Assume the diodes of the display are low level signals!

B) Deduce And Simplify the logic expressions for a, b, c, d, e, f and g.

0 ⇒ a·b·C·d·e·f 1 ⇒ b·c 2 ⇒ a·b·d·e·g 3 ⇒ a·b·c·d·g

4 ⇒ b·c·f·g 5 ⇒ a·c·d·f·g 6 ⇒ a·c·d·e·f·g 7 ⇒ a·b·c

8 ⇒ a·b·c·d·e·f·g 9 ⇒ a·b·c·d·f·g

$a = \sum_m (0,2,3,5,6,7,8,9) + \sum_x (10,11,12,13,14,15)$

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

They said assume low level, so, ↳

$a \Rightarrow A + C + B \cdot D + \bar{B} \cdot \bar{D} \Rightarrow \bar{a} = B \cdot \bar{C} \cdot \bar{D} + \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot D$

$\bar{b} = \sum_m (0,1,2,3,4,7,8,9) + \sum_x (10,11,12,13,14,15)$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$b \Rightarrow \bar{B} + C \cdot D + \bar{C} \cdot \bar{D}$

Low level $\bar{b} \Rightarrow B\bar{C} \cdot \bar{D} + B \cdot \bar{C} \cdot D$

$\bar{c} = \sum_m (0,1,3,4,5,6,7,8,9) + \sum_x (10,11,12,13,14,15)$

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$C \Rightarrow B + \bar{C} + D$

$\bar{C} = \overline{B + \bar{C} + D} \Rightarrow \bar{B} \cdot C \cdot \bar{D}$ ↳

$d = \sum m(0,2,3,5,6,8,9) + \sum x(10,11,12,13,14,15)$

d= AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	X	X	X	X
10	1	1	X	X

$d \Rightarrow A + \bar{B}C + \bar{B}\bar{D} + C\bar{D} + B\bar{C}D$

$\bar{d} = \overline{A + \bar{B}C + \bar{B}\bar{D} + C\bar{D} + B\bar{C}D}$

$\bar{d} \Rightarrow B\bar{C}D + B\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D$

$e \Rightarrow \sum m(0,2,5,6,8) + \sum m(10,11,12,13,14,15)$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	X	X	X	X
10	1	0	X	X

$e \Rightarrow \bar{B}\bar{D} + C\bar{D}$

$\bar{e} \Rightarrow \overline{\bar{B}\bar{D} + C\bar{D}}$

$\bar{e} \Rightarrow \overline{\bar{B}\bar{D}} \cdot \overline{C\bar{D}} \Rightarrow B\bar{C} + BD + \bar{C}D + D$

$\bar{e} \Rightarrow D + B\bar{C}$

$f \Rightarrow \sum m(0,4,5,6,8,9) + \sum x(10,11,12,13,14,15)$

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$f \Rightarrow A + B\cdot\bar{C} + B\cdot\bar{D} + \bar{C}\cdot\bar{D}$

$\bar{f} = \overline{A + B\cdot\bar{C} + B\bar{D} + \bar{C}\cdot\bar{D}}$

$\bar{f} \Rightarrow \bar{B}\cdot C + C\cdot D + \bar{A}\cdot\bar{B}\cdot D$

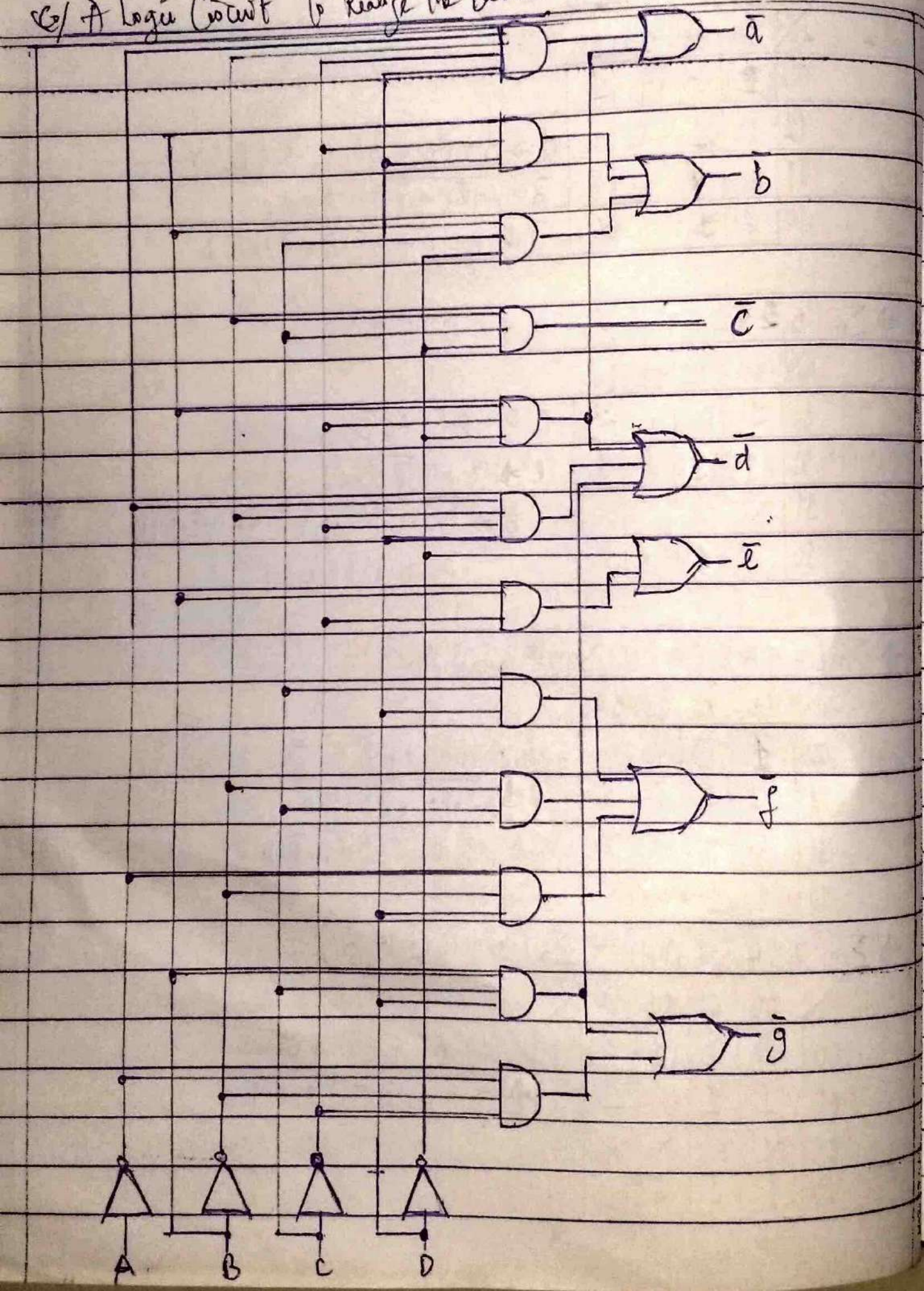
$g \Rightarrow \sum m(2,3,4,5,6,8,9) + \sum x(10,11,13,13,14,15)$

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$g \Rightarrow A + D\bar{C} + \bar{B}C + C\bar{D}$

$\bar{g} = \bar{A}\cdot\bar{B}\cdot\bar{C} + B\cdot C\cdot D$

Q/ A Logic Circuit To Realize the Decoder



Exercise 3.16 ⇒ HEX-to-7-Segment Decoder

Hex-to-7-Segment

Hex ⇒ 0-9; A-F

Hex ⇒ 0-F

Meaning 0-15 (16 symbol)

Truth Table for the Decoder with Low Level Signals:

Symbol	A	B	C	D	$\bar{a}$	$\bar{b}$	$\bar{c}$	$\bar{d}$	$\bar{e}$	$\bar{f}$	$\bar{g}$
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0
A	1	0	1	0	0	0	0	1	0	0	0
b	1	0	1	1	1	1	0	0	0	0	0
C	1	1	0	0	0	1	1	0	0	0	1
d	1	1	0	1	1	0	0	0	0	1	0
e	1	1	1	0	0	0	1	0	0	0	0
F	1	1	1	1	0	1	1	1	0	0	0

[Truth table gotten from Fig 3.70 (a) & (b)]

[The output is basically the same as BCD to 7 Segment for the 0-9 Symbol but inverted in this condition]

OR

[Cut the Symbol equivalent and inverse]

(b) Getting And Simplifying logic expressions for $\bar{a}, \bar{b}, \bar{c}, \bar{d}, \bar{e}, \bar{f}, \bar{g}$

$\bar{a} = \Sigma m(1, 4, b, d)$ in $\Sigma m(1, 4, 11, 13)$

AB \ CD	00	01	11	10
00	0	(1)	0	0
01	(1)	0	0	0
11	0	(1)	0	0
10	0	0	(1)	0

$\bar{a} = \bar{A}\cdot\bar{B}\cdot\bar{C}\cdot D + \bar{A}\cdot B\cdot\bar{C}\cdot\bar{D} + A\cdot B\cdot\bar{C}\cdot D + A\cdot\bar{B}\cdot C\cdot D$

$\bar{b} = \Sigma m(3, 6, b, C, F)$ $\quad b \Rightarrow 11$; $C = 12$; $F = 15$

AB \ CD	00	01	11	10
00	0	0	0	0
01	(0)	(1)	0	(1)
11	(1)	0	(1)	0
10	0	0	(1)	0

$\bar{b} \Rightarrow A\cdot\bar{B}\cdot D + \bar{A}\cdot B\cdot\bar{C}\cdot D + A\cdot B\cdot C\cdot\bar{D} + A\cdot\bar{B}\cdot\bar{C}\cdot\bar{D}$

$\bar{c} = \Sigma m(2, C, e, f)$ Binary $\Rightarrow \Sigma m(2, 12, 14, 15)$

AB \ CD	00	01	11	10
00	0	0	0	(1)
01	0	0	0	0
11	(1)	0	(1)	(1)
10	0	0	0	0

$\bar{c} \Rightarrow A\cdot B\cdot C + A\cdot B\cdot\bar{D} + \bar{A}\cdot\bar{B}\cdot C\cdot\bar{D}$

$d = \sum m(1,4,7,A,F)$ Binary $\Rightarrow \sum m(1,4,7,10,15)$

AB \ CD	00	01	11	10
00	0	1	0	0
01	1	0	1	0
11	0	0	1	0
10	0	0	0	1

$d \Rightarrow B\cdot C\cdot D + \bar{A}\cdot\bar{B}\cdot\bar{C}\cdot D + \bar{A}\cdot B\cdot\bar{C}\cdot\bar{D} + A\cdot\bar{B}\cdot C\cdot\bar{D}$

$\bar{e} = \sum m(1,3,4,5,7,9)$

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	0
11	0	0	0	0
10	0	1	0	0

$\bar{e} = \bar{A}\cdot C\cdot D + \bar{A}\cdot B\cdot\bar{C} + \bar{B}\cdot C\cdot D$

$\bar{f} = \sum m(1,2,3,7,d)$; $d = 13$

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	1	0
11	0	1	0	0
10	0	0	0	0

$f \Rightarrow \bar{A}\cdot\bar{B}\cdot D + \bar{A}\cdot C\cdot D + \bar{A}\cdot\bar{B}\cdot C + A\cdot B\cdot\bar{C}\cdot D$

~~$\bar{A}\cdot D + \bar{A}\cdot\bar{B}$~~

$\bar{g} = \sum m(0,1,7,C)$; C = 12

AB \ CD	00	01	11	10
00	1	1	0	0
01	0	0	1	0
11	1	0	0	0
10	0	0	0	0

$\bar{g} \Rightarrow \bar{A}\cdot\bar{B}\cdot\bar{C} + A\bar{B}\bar{C}\cdot\bar{D} + \bar{A}BCD$

Symbol	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0
A	1	0	1	0	0	0	0	1	0	0	0
b	1	0	1	1	1	1	0	0	0	0	0
C	1	1	0	0	0	1	1	0	0	0	1
d	1	1	0	1	1	0	0	0	0	1	0
e	1	1	1	0	0	0	1	0	0	0	0
F	1	1	1	1	0	1	1	1	0	0	0

Truth Table of the HEX-to-7-Segment decoder.

Exercise 3.17 – Analysis of Logic Circuit

(a) Analysis of Figure 3.71

To determine the function;

Barrel shifters shift data by a certain number of positions to the left or the right.

Diagram indicates, as a 1-bit barrel shifter.

input → $X_4, X_3, X_2, X_1, X_0, X_{n-1}$

output → Y_3, Y_2, Y_1, Y_0

enable signal E;

Signal D. if =1 shift to the left

E	S	D	Y_3	Y_2	Y_1	Y_0	
0	x	x	0	0	0	0	← Enable signal E=0; it doesn't
1	0	x	X_3	X_2	X_1	X_0	matter about other input have
1	1	0	X_4	X_3	X_2	X_1	← Enable signal E=1; S=0
1	1	1	X_3	X_2	X_1	X_{-1}	∴ D's input does not matter.

Barrel Shifter regular truth table

Confirming the truth table from the fig 3.7

$Y_0 = X_1 \cdot E \cdot S \cdot D + E \cdot X_0 \cdot \bar{S} + E \cdot X_1 \cdot S \cdot \bar{D}$

$Y_0 = E[X_1 \cdot S \cdot D + X_0 \cdot \bar{S} + X_1 \cdot S \cdot \bar{D}]$ @ E=1; S=1 D=1

$Y_0 = 1[X_1 \cdot 1 \cdot 1 + X_0 \cdot 0 + X_1 \cdot S \cdot 0]$

$Y_0 = X_1$

Y_1 → D = 1 for a shift to the left

0 for a shift to the right

B

Control Signals			Data Outputs							
S_2	S_1	S_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	1	0	D_7	D_6	D_5	D_4	D_3	D_2	D_1
0	1	0	0	0	D_7	D_6	D_5	D_4	D_3	D_2
0	1	1	0	0	0	D_7	D_6	D_5	D_4	D_3
1	0	0	0	0	0	0	D_7	D_6	D_5	D_4
1	0	1	0	0	0	0	0	D_7	D_6	D_5
1	1	0	0	0	0	0	0	0	D_7	D_6
1	1	1	0	0	0	0	0	0	0	D_7

Truth table for 2:1 multiplexer → Ex 3.17 (b)

(b) Analysing Fig 8.72 based on 2:1 multiplexers

Data inputs are ; $D_7, D_6, D_5, D_4, D_3, D_2, D_1, D_0$

Data Outputs ; $Y_7, Y_6, Y_5, Y_4, Y_3, Y_2, Y_1, Y_0$

Control signals ; S_2, S_1, S_0

Logic equations ; $Y_7 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_7$

$Y_6 \Rightarrow (\bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_6 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_7)$

$Y_5 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_5 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_6 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_7$

} Logic equations were derived from Fig 8.72 Logic circuit 2

$Y_4 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_4 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_5 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_6 + \bar{S}_2 \cdot S_1 \cdot S_0 \cdot D_7$

$Y_3 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_3 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_4 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_5 + \bar{S}_2 \cdot S_1 \cdot S_0 \cdot D_6 + S_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_7$

$Y_2 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_2 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_3 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_4 + \bar{S}_2 \cdot S_1 \cdot S_0 \cdot D_5 + S_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_6 + S_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_7$

$Y_1 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_1 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_2 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_3 + \bar{S}_2 \cdot S_1 \cdot S_0 \cdot D_4 + S_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_5 + S_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_6 + S_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_7$

$Y_0 \Rightarrow \bar{S}_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_0 + \bar{S}_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_1 + \bar{S}_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_2 + \bar{S}_2 \cdot S_1 \cdot S_0 \cdot D_3 + S_2 \cdot \bar{S}_1 \cdot \bar{S}_0 \cdot D_4 + S_2 \cdot \bar{S}_1 \cdot S_0 \cdot D_5 + S_2 \cdot S_1 \cdot \bar{S}_0 \cdot D_6 + S_2 \cdot S_1 \cdot S_0 \cdot D_7$

CHAPTER 4 ⟨Chapter 5 Flip Flops⟩

1.10 Exercises

EXERCISE 1-1 → Propose an equivalent switch-based circuit for each of the circuits in Figure 1.37

What is the function of these circuits?

It is a switch bouncer.

↳ When $\bar{R}=0$, the output Q is set to 1 when the signal $\bar{S}$ goes HIGH for the first time.

↳ $\bar{R}=0$ $Q=1$ $\bar{S}=1$

$\bar{R}=1$ $Q=0$ $\bar{S}=0$ → when signal was turned off ⟨0⟩ the output Q resets and $\bar{R}=1$. This shows it's a switch bouncer.

EXERCISE 1-2 → Consider the T-latch in Figure 1.38

Determine the characteristics equations of this latch.

If latches can be gotten from JKT or SR latches.

With the diagram from the diagram we see a SR latch.

For SR latch is; $Q^+ = \bar{S}\cdot C + (\bar{R}+\bar{C})\cdot Q$

from the diagram;

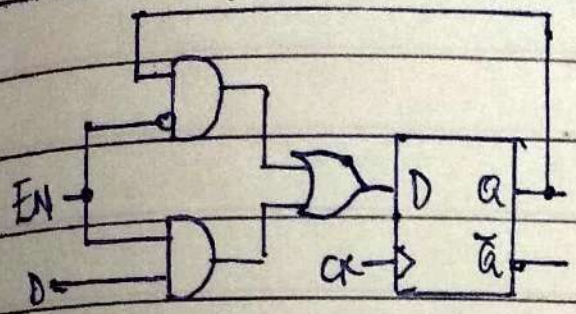
$S=\bar{Q}$ $R=Q$ $C=T$

↳ $Q^+ = \bar{Q}\cdot\bar{Q}\cdot T + (Q+\bar{T})\cdot Q$

$Q^+ \Rightarrow \bar{Q}\cdot T + \bar{T}\cdot Q$

$Q^+ \Rightarrow T \oplus Q$

Exercise 1.3 ⇒ Analyze and construct the truth table for the flip-flop shown in Fig 1.39



The Gate shows that it a 2:1 multiplexer connected for a D latch.

$Q^+ \Rightarrow D \cdot EN + \overline{EN} \cdot Q$

EN	D	CK	Q^+	$\overline{Q^+}$
X	X	0	Q	$\overline{Q}$
X	X	1	Q	$\overline{Q}$
0	X	↓	Q	$\overline{Q}$
1	0	↓	0	1
1	0	↓	1	0

Exercise 1.4 ⇒ Positive Edge-Triggered D Flip-Flop in Fig 1.40(a)

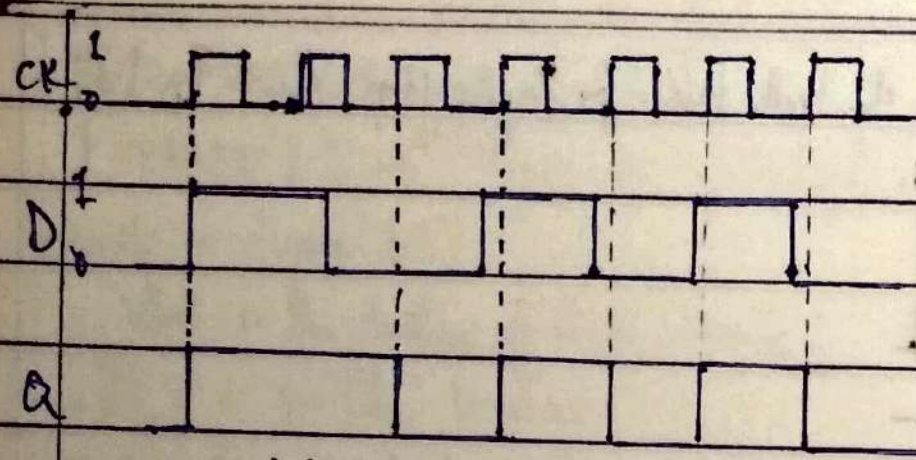
A positive edge trigger is when the clock changes from 0-1 and that is when the flip flop is activated. Ls (↑)

A triggered flip flop is also known as a clocked flip-flop.

Sometimes you'll hear Positive Going Transition (PGT) and Negative Going Transition (NGT). PGT ⇒ ↑ & NGT ⇒ ↓

Also we, falling edge and rising edge.

So the output Q takes the logic state D when the clock changes from 0 to 1.



Completed timing-diagram in Figure 1.10(b)

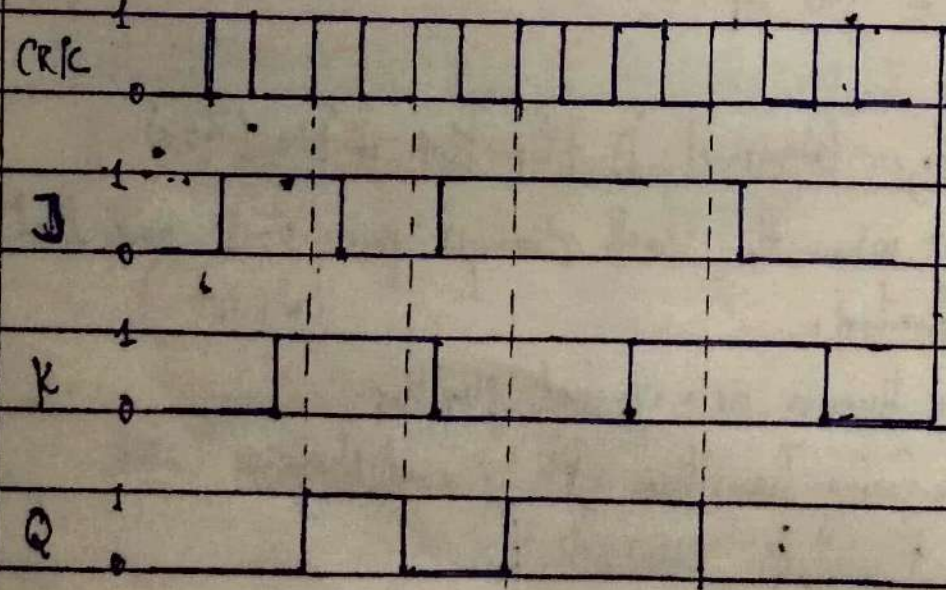
Exercise 1.5 ; A Positive edge-triggered JK Flip Flop

When the clock turns from 0→1

JK equation $Q^+ = J\bar{Q}\cdot C + (\bar{K}+\bar{C})\cdot Q$

@ $C=1$ → $J\cdot\bar{Q} + \bar{K}\cdot Q \Rightarrow J \oplus Q$

@ $C=0$ $\quad Q^+ = Q$

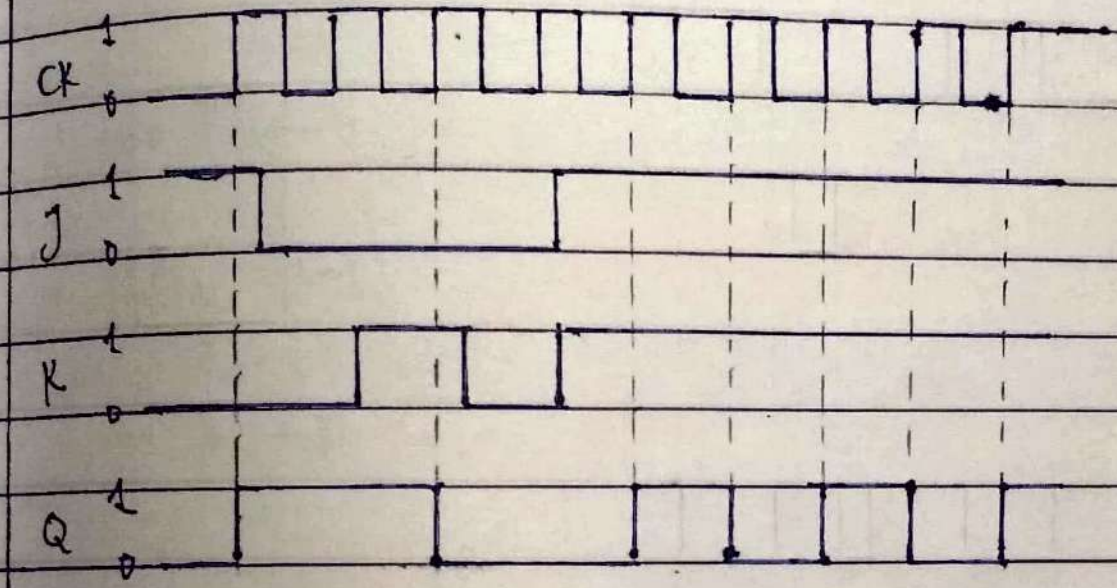


J	K	CK	Q^+
0	0	↓	Q
0	1	↓	0
1	0	↓	1
1	1	↓	$\bar{Q}$

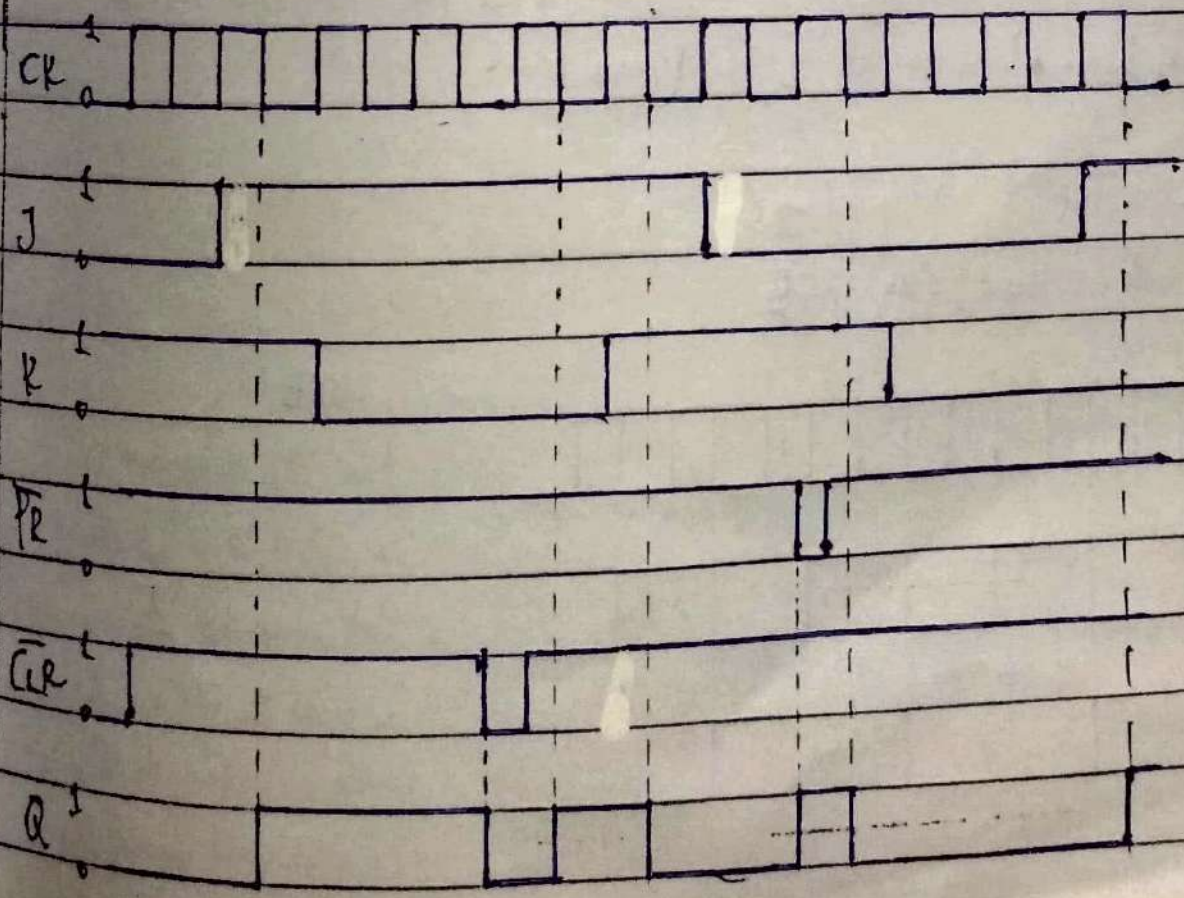
Exercise 1.6: Master-Slave JK Flip-Flop

In Figure 1.42(a)

Complete timing diagram in 1.42(b).

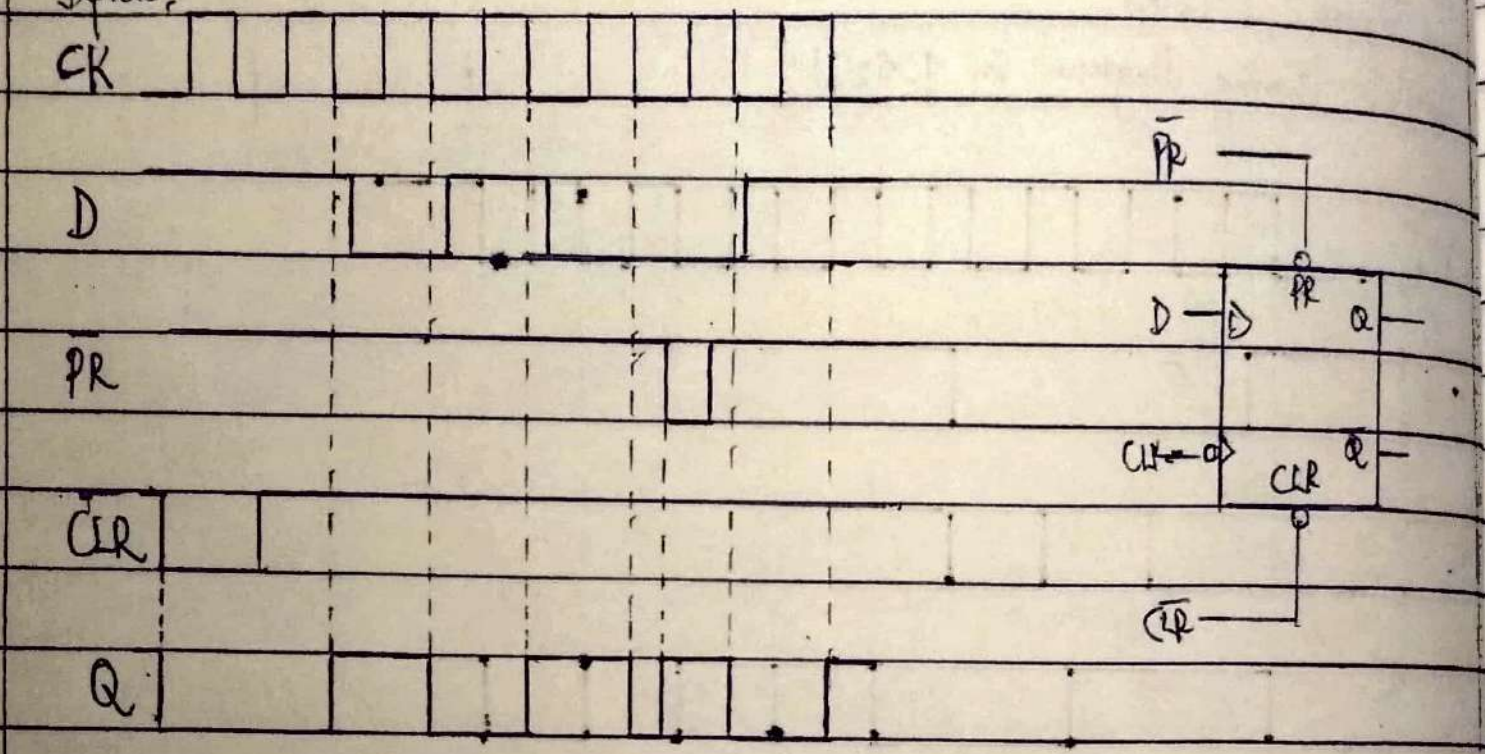


Exercise 1.7: J-K Flip Flop with Asynchronous Inputs.... Complete the timing diagram in Figure 1.43(b)



Exercise 1.8 :- Logic Circuit for A D Flip-Flop With Asynchronous

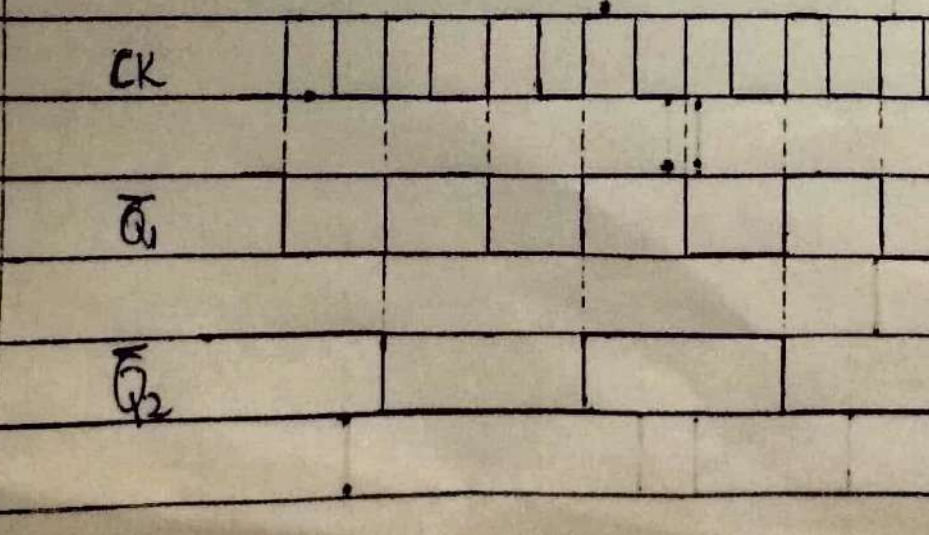
Inputs:



The latch activates on the falling edge of CK, but also because it is asynchronous, the latch is also affected by the Clear (CLR) and PR.
Once the clear remains at one (HIGH) having the pulse from $\overline{PR}$; D is high at $\downarrow$ Q becomes high, D is low, at $\downarrow$ Q becomes low.

Exercise 1.9:- Two D Flip Flops

(a) D latch with an input; Fig 1.45

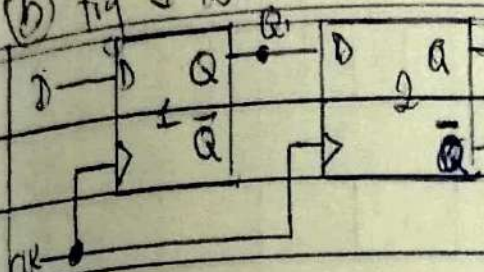


NOTE!

From the diagram we see $\overline{Q_1}$ is high but changes state when ck $\downarrow$

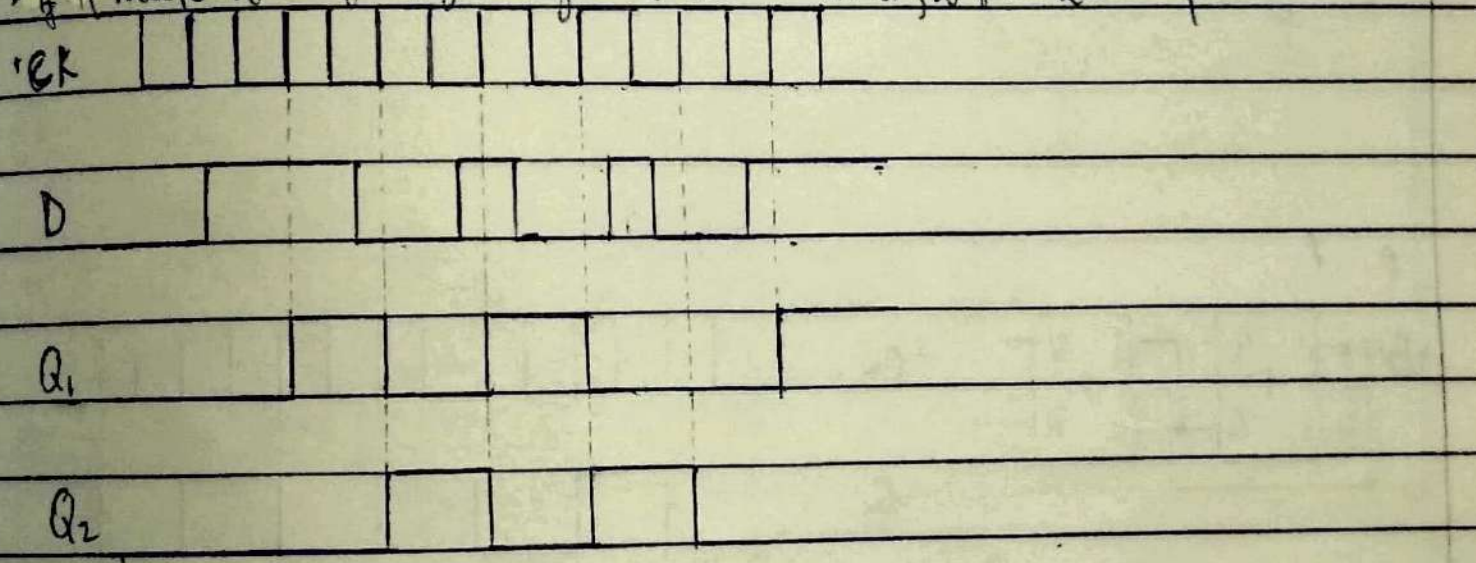
Q_1 takes the role of CK in the 2nd ding, so Q_2 changes state when $\downarrow Q_1$

(b) fig 1.46

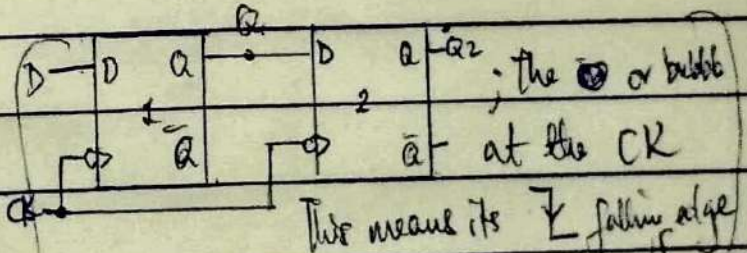


From the Diagram D has its own input pulse along with the CLK. Q_1 activates as high when D is high and CLK ↓
Q_2 activates to low (0) when D is low and CLK ↓

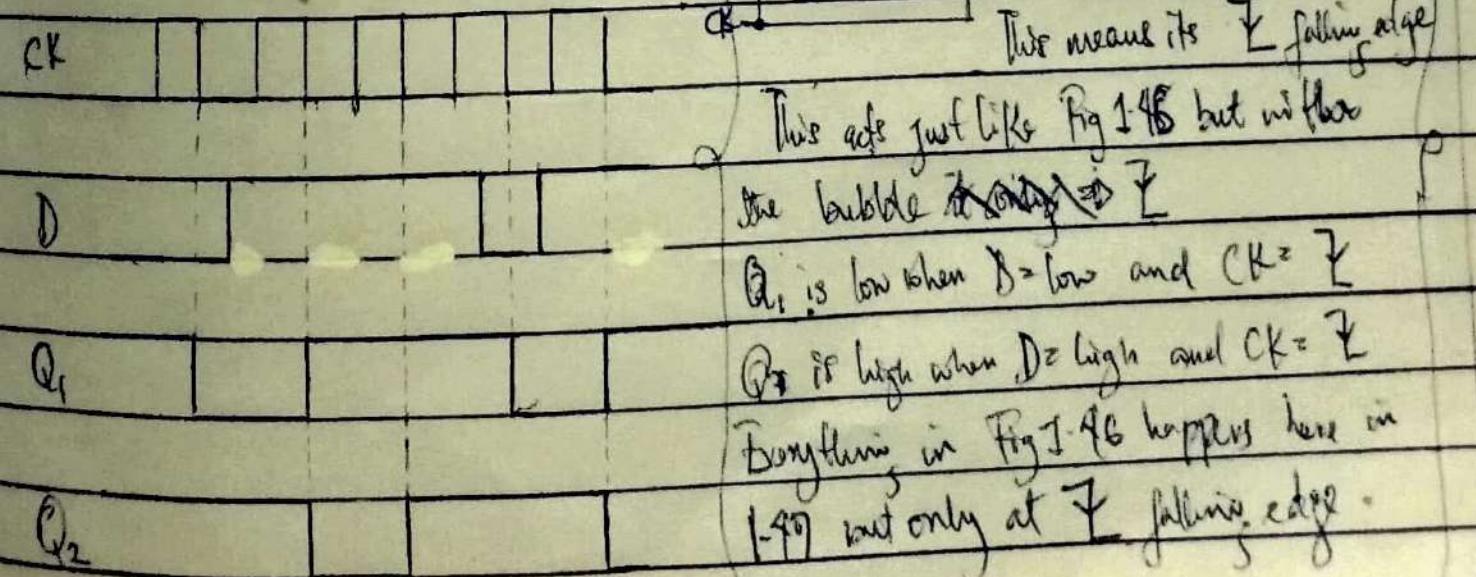
for Q_2;
Q_2 now works as Q_1 using the input pulse and CLK as the activator. Hence;
Q_1 has a propagation delay hence, Q_2 takes Q_1 pulse just be fore it changes
→ if it meets it at low right before it becomes HIGH, it takes LOW & vice-versa



(c) Fig 1.47



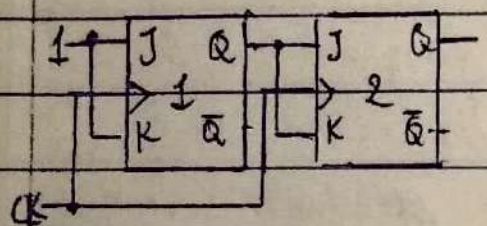
; the ○ or bubble at the CK
This means its ↓ falling edge

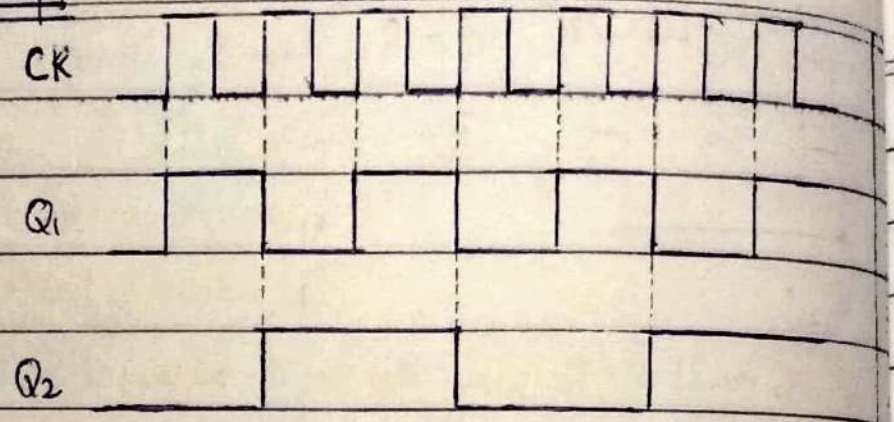


This acts just like Fig 1.46 but with the bubble ⇒ ↓
Q_1 is low when D = low and CK = ↓
Q_2 is high when D = high and CK = ↓
Everything in Fig 1.46 happens here in 1.47 but only at ↓ falling edge.

Exercise 1.10:- JK Flip Flops

Fig 1.48

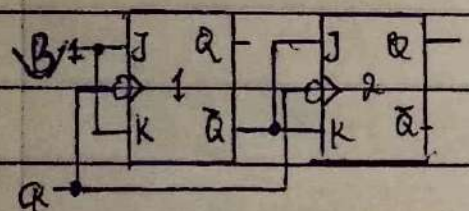


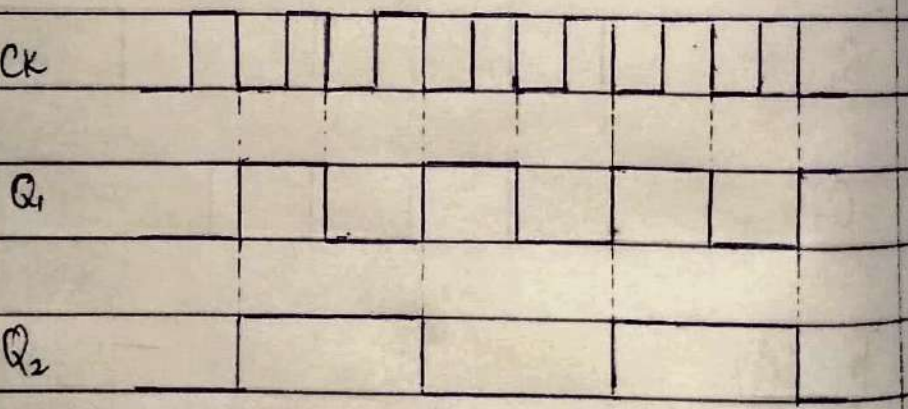


Completed timing diagram

$Q^t = J \cdot Q + \bar{K} \cdot Q$

P





Complete timing diagram

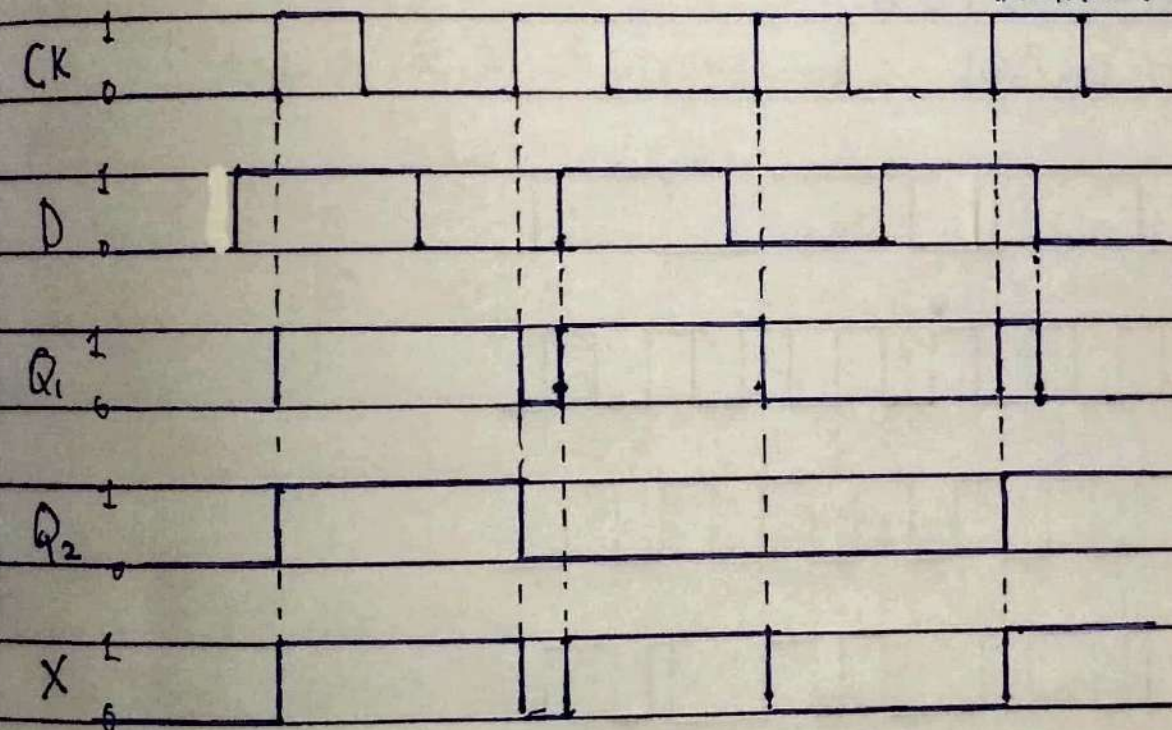
Exercise 1.11 → Two D Flip-Flops And A Combinational Logic Circuit

From figure 1-50(a), we see that both Q_1 and Q_2 are edge triggered, but Q_2 really is more edge triggered as Q_1 only activates whatever / assumes whatever D is while the CK is high.

Meaning if D's pulse changes from a high to low while the CK period is ongoing Q_1 will also change to low, but Q_2 won't.

If this state persist till the next PGT {↓} Q_2 will change to LOW while Q_1 would already have change, hence will remain LOW!

↳ PROPAGATION DELAY!

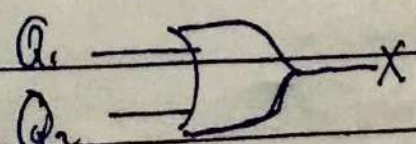


we can see from the diagram;

$X = Q_1 + Q_2$ ⟨OR Gate⟩

B/ It can be implemented with an OR gate

Q_1	Q_2	X
0	0	0
0	1	1
1	0	1
1	1	1



Exercise 1.12

Determine the characteristic equation for each of the synchronous D flip-flops in Fig 1.51

a)

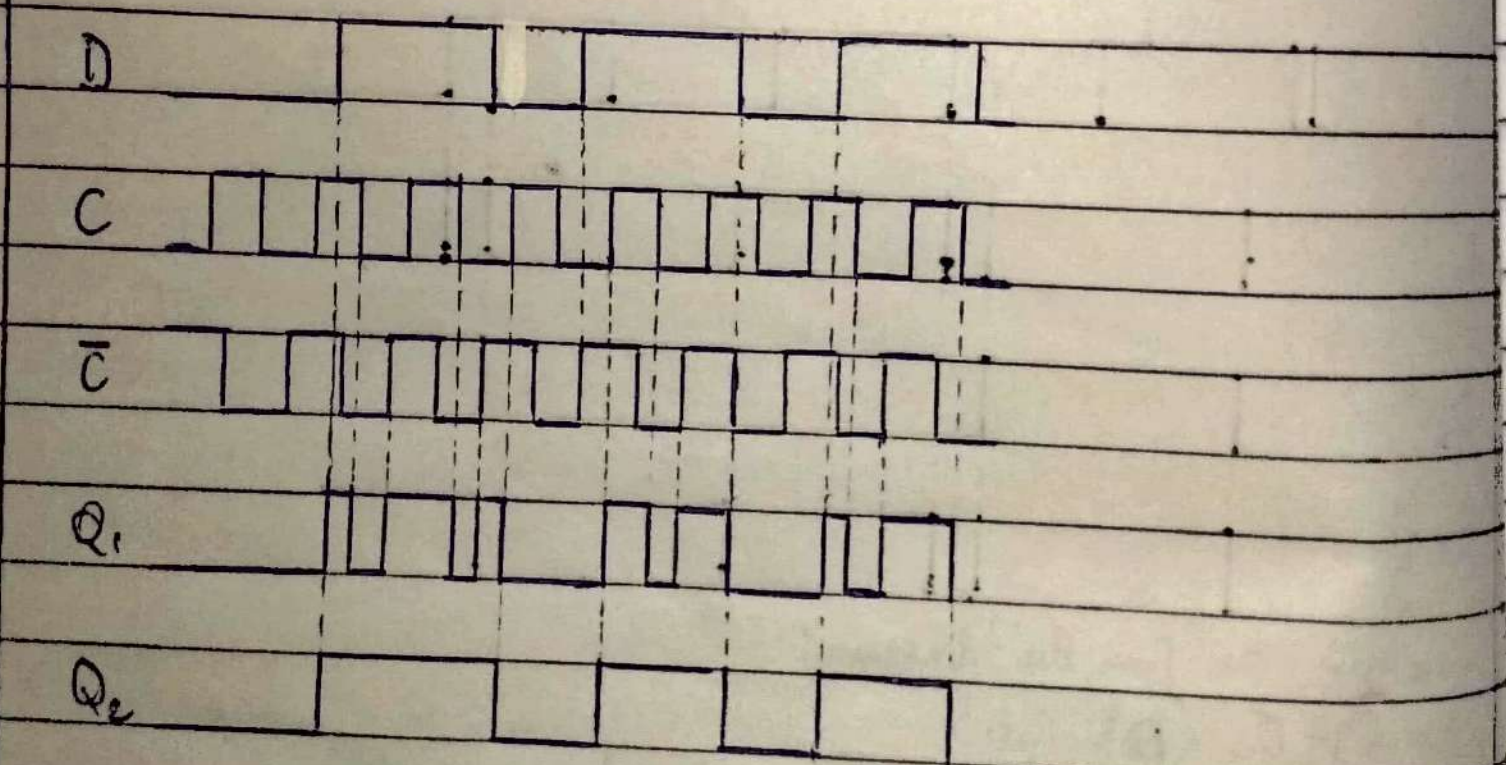
Latch L1 from Fig 1.51 (a)

$Q^+ = C \cdot D + \bar{C} \cdot Q$

Latch L2 from Fig 1.51 (b)

$\hookrightarrow Q^+ = C \cdot D + D \cdot Q + \bar{C} \cdot Q$

So from this equations, we can complete the timing diagram



Propagation delay used to generate $\bar{C}$

If $D = 1 \Rightarrow Q = 1$

Latch 1 becomes $\Rightarrow C + \bar{C}$

Latch 2 becomes $\Rightarrow C + 1 + \bar{C} = 1$

Hence Latch Q_2 operates correctly!

CS/Q	0	1
00	0	(1)
01	0	(1)
11	(1)	(1)
10	0	0

CS/Q	0	1
00	0	(1)
01	0	(1)
11	(1)	(1)
10	0	0

→ Delay FF or Synchronous ??

Exercise 1.13 ⇒ Converting Between Different Flip-Flops.

on Figure 1.53

(a) Equivalence of Fig 1.53(a) ⇒ D Flip Flop & JK Flip Flop

$Q^+ = D = T \oplus Q$

JK Flip Flop in JK flip flop, we have;

$Q^+ = J\bar{Q} + \bar{K}Q$ ⇒ D

$J = T$ and $K = \bar{T}$; $Q^+ \Rightarrow T\bar{Q} + \bar{T}Q \Rightarrow T \oplus Q$

(b) Fig 1.53(b) JK Flip Flop:

$Q^+ = D = J\bar{Q} + \bar{K}Q$

(c) Fig 1.53(c)

The first circuit, JK based - we get

$J = D$ & $K = \bar{D}$

Hence the characteristic of the flip flop is; $Q^+ = J\bar{Q} + \bar{K}Q \langle \Rightarrow D \rangle$

(d) Fig 1.53(d)

Expression for the 2:1 multiplexer D flip flop

is $Q^+ \Rightarrow D\,Q\cdot\overline{EN} + D\cdot EN$

as an input into the J-K flip flop which is $\langle J\bar{Q} + \bar{K}Q \rangle$

$J \Rightarrow D\,D\cdot EN \quad K \Rightarrow \bar{D}\cdot EN$

Q^t becomes

$Q^t = [D \cdot EN] \cdot \bar{Q} + [\bar{D} \cdot EN] \cdot Q$

$D \cdot EN \cdot \bar{Q} + (D + \overline{EN}) \cdot Q$

$D \cdot EN \cdot \bar{Q} + D \cdot Q + \overline{EN} \cdot Q$

$D(EN \cdot \bar{Q} + Q) + \overline{EN} \cdot Q$

$D \cdot Q + D \cdot EN + \overline{EN} \cdot Q$

CONSENSUS THEOREM

$DEN + Q \cdot \overline{EN}$

Q^+ becomes

$Q^+ = [D \cdot EN] \cdot \bar{Q} + [\overline{D \cdot EN}] \cdot Q$

$\hookrightarrow D \cdot EN \cdot \bar{Q} + (D + \overline{EN}) \cdot Q$

$D \cdot EN \cdot \bar{Q} + D \cdot Q + \overline{EN} \cdot Q$

$D(EN \cdot \bar{Q} + Q) + \overline{EN} \cdot Q$

$D \cdot Q + D \cdot EN + \overline{EN} \cdot Q$ } CONSENSUS THEOREM

$\hookrightarrow DEN + Q \cdot \overline{EN}$

DIGITAL INTEGRATED CIRCUIT TECHNOLOGY

CHAPTER 5

EXERCISE 5.1

(a) q $\langle V_{IH}, V_{IL}, V_{OH}, V_{OL} \rangle$

b. γ

c. β

d. α

e. α

f. β

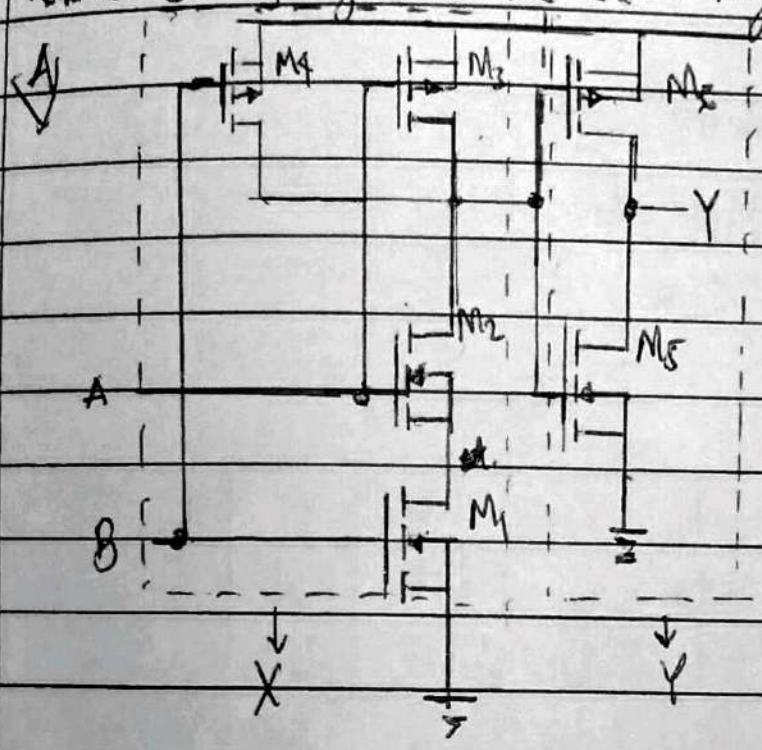
g. γ

h. β

i. γ

j. α

Exercise 5.2 ; Figure 5-17 (a) What logic gate is being implemented??



The Part X has two N channel transistors connected in Series with two P channel transistors connected in Parallel hence it is A NAND gate.
The Part Y is an Inverter. When the input signal takes the high level, the M_5 transistor conducts while M_2 is off and out put takes Low.
With (NAND) connected to INVERTER the whole circuit becomes an AND Gate { with Low = 0, High = 1 }

A	B	M_1	M_2	M_3	M_4	M_5	M_6	Y	
0	0	OFF	OFF	ON	ON	ON	OFF	LOW	→ 0
0	1	ON	OFF	ON	OFF	ON	OFF	LOW	→ 0
1	0	OFF	ON	OFF	ON	ON	OFF	LOW	→ 0
1	1	ON	ON	OFF	OFF	OFF	ON	High	→ 1

* I prefer 0 & 1 → It is Logic ....

EXERCISE 5.3

(b) Fig 5-17(b)

Two P channel transistors connected in Series with two N-Channel transistors connected in Parallel ⇒ NOR gate.

NOR gate connected to an INVERTER

Fig 5-17(b) is an OR GATE

A	B	N_1	N_2	N_3	N_4	N_5	N_6	Y
0	0	OFF	OFF	ON	ON	ON	OFF	LOW
0	1	ON	OFF	OFF	ON	OFF	ON	HIGH
1	0	OFF	ON	ON	OFF	OFF	ON	HIGH
1	1	ON	ON	OFF	OFF	OFF	ON	HIGH

N.B → There is another mode of analysis. You break the circuit up get the states of the transistor then create a truth table, invert the output Y. Same can be done for A (Fig 5.17A)

Exa

EXERCISE 5.3

(a) Fig 5.18(a)

$E = A\bar{B}\bar{C}$

(b) Fig 5.18(b)

$F \Rightarrow \overline{A \cdot B} \cdot \overline{C \cdot D}$

$\hookrightarrow (\bar{A}+\bar{B}) \cdot (\bar{C}+D)$ OR $\bar{A}\cdot C + \bar{A}D + B\bar{C} + B\bar{D}$

(c) Fig 5.18(c)

$G \Rightarrow \overline{\bar{A} \cdot B\bar{D}}$

$\Rightarrow A + \bar{B} + D$

EXERCISE 5.4

$P_D = V_{cc} I_{cc}$ $\hookrightarrow I_{cc} \Rightarrow \left(\frac{I_{cc_H} + I_{cc_L}}{2}\right)$

$P_D = V$

$I_{cc} \Rightarrow \left(\frac{1.5 + 4.5}{2}\right) mA$

$I_{cc} \Rightarrow 3mA \qquad V_{cc} = 5V$

$\therefore P_D = 5 \times 3mA$

$P_D \Rightarrow 15mW.$

(b) What is the highest possible frequency (f) if the maximum value of the power, $P_D \Rightarrow 15mW$

$C_L = 10pF$

$C_{PD} = 22pF$, $V_{DD} = 3V$ $I_{DD} = 20\mu A$

$$P_D = (C_L + C_{PD}) V_{DD}^2 f + V_{DD} I_{DD}$$

$$f \Rightarrow \frac{P_D - V_{DD} I_{DD}}{(C_L + C_{PD}) V_{DD}^2}$$

Maximum frequency (f) $\Rightarrow \frac{15\times10^{-3} - 3(20\times10^{-6})}{(10\times10^{-12} + 22\times10^{-12}) \times 3^2} \Rightarrow \frac{15\times10^{-3} - 60\times10^{-6}}{30\times10^{-12}\times3^2}$

$\hookrightarrow 55.3 MHz$